Methods in Human Geography

Quantitative Methods: Spatial Analysis I



Justin van Dijk
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This week

Part I

- Spatial data.
- GIScience and GISystems.

Part II

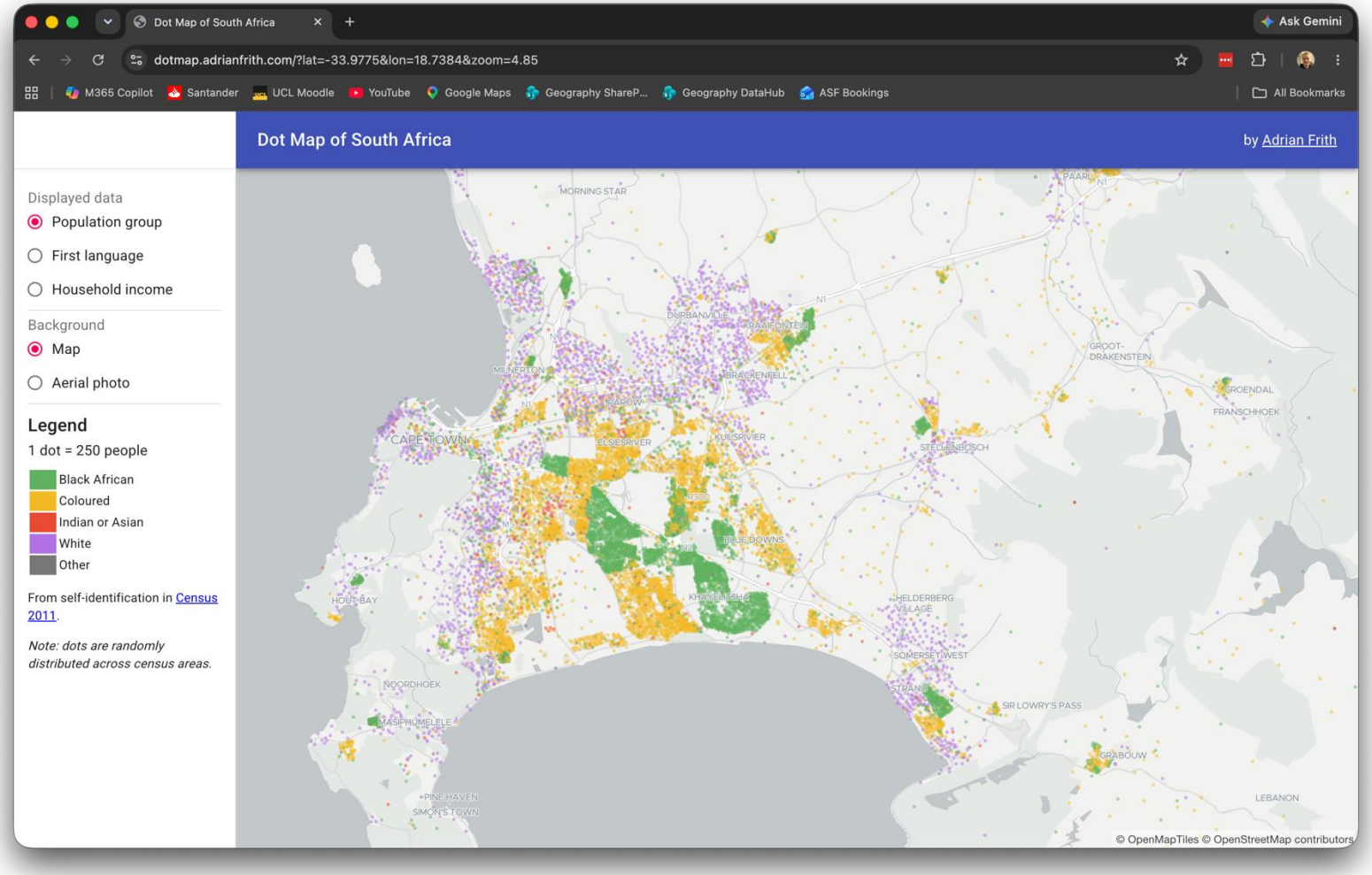
- Data sources.
- Attribute joins.

Spatial data

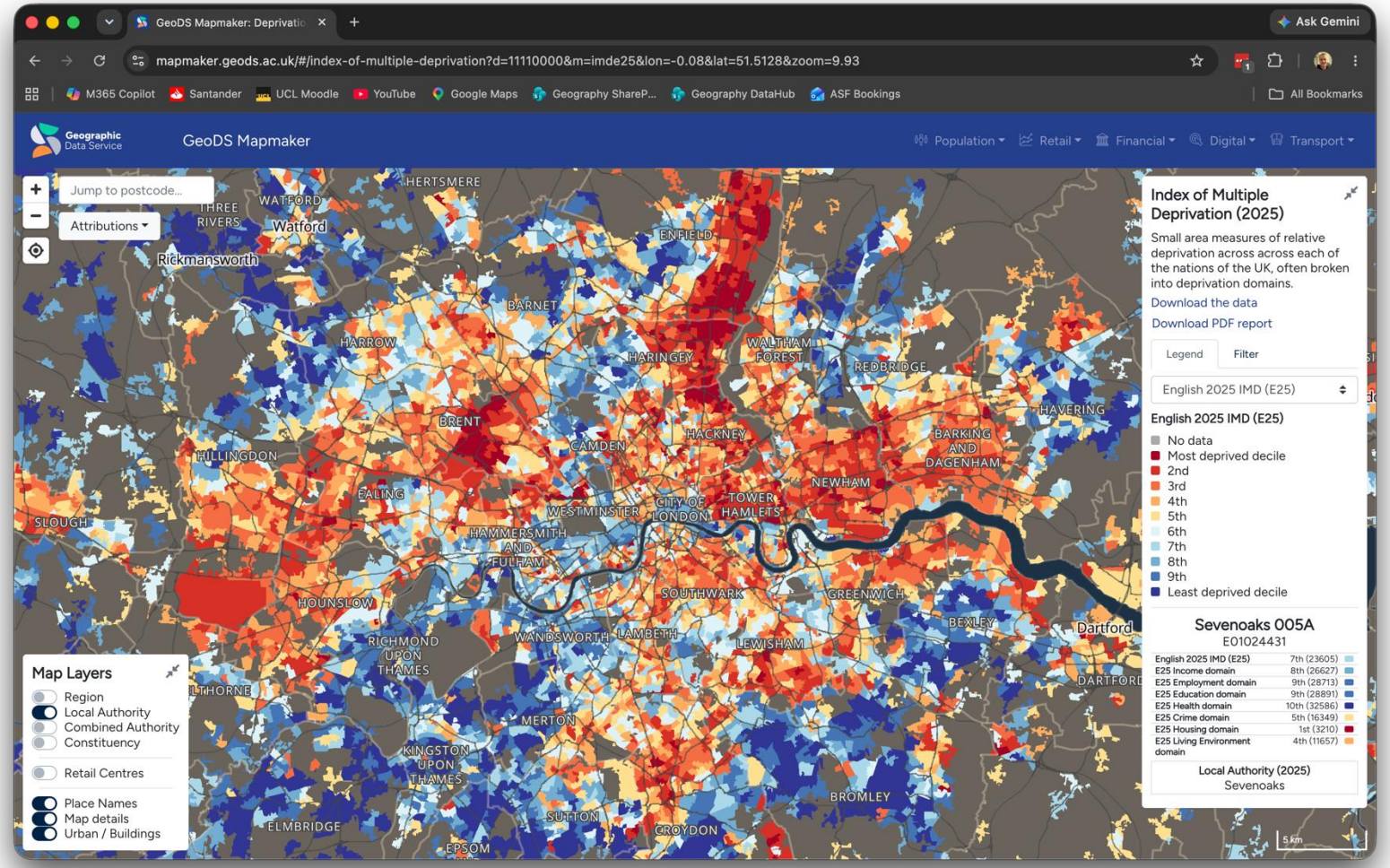
Why is spatial data important?

- Many challenges and problems are inherently geographical.
- Almost all data is tied to specific locations and varies across space.
- Spatial data helps us answer the question: what is this place like compared to other places?
- Maps present spatial data in a format that the human brain understands more intuitively.

Segregation in Cape Town



Index of Multiple Deprivation in England

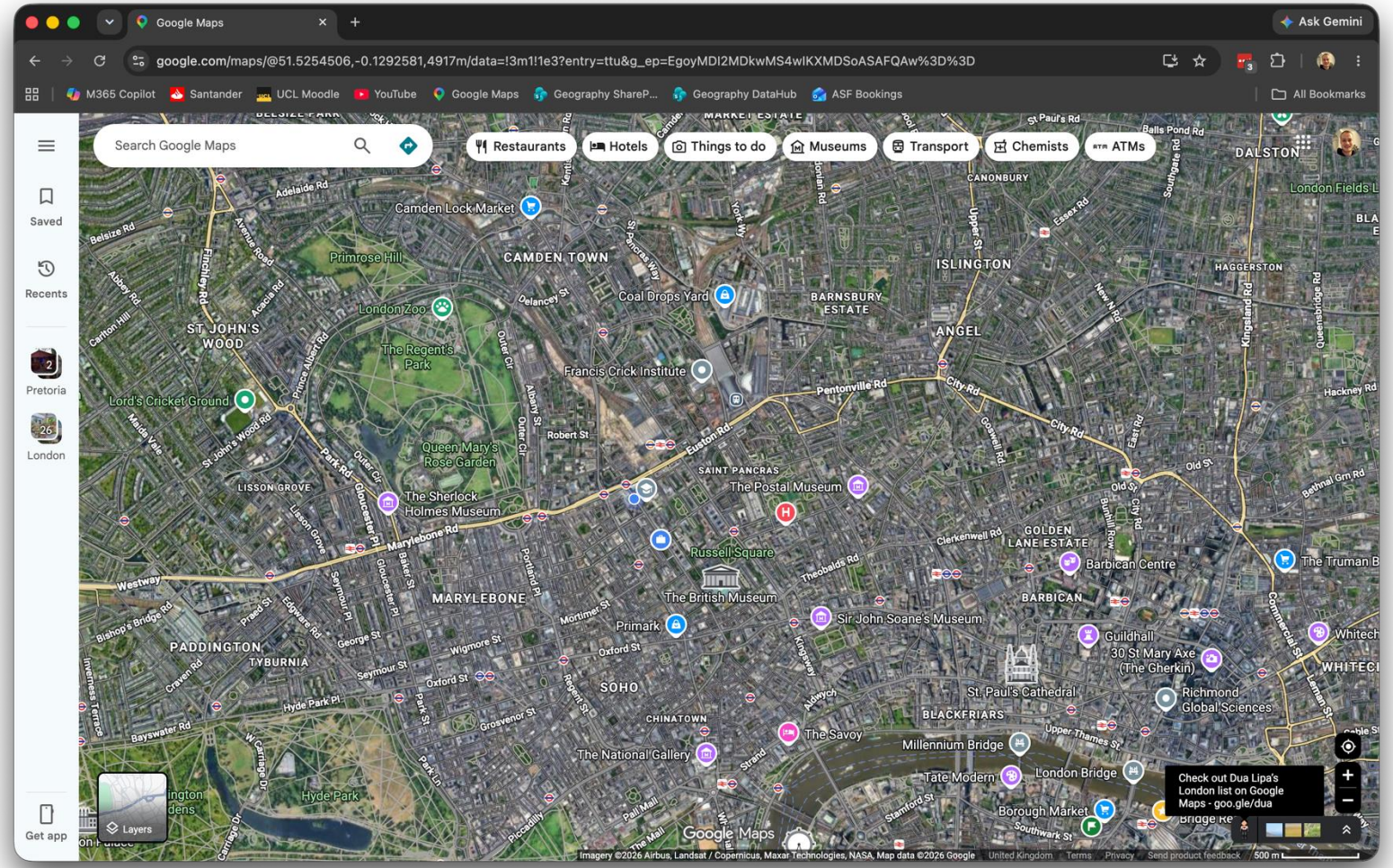


GIScience and GISystems

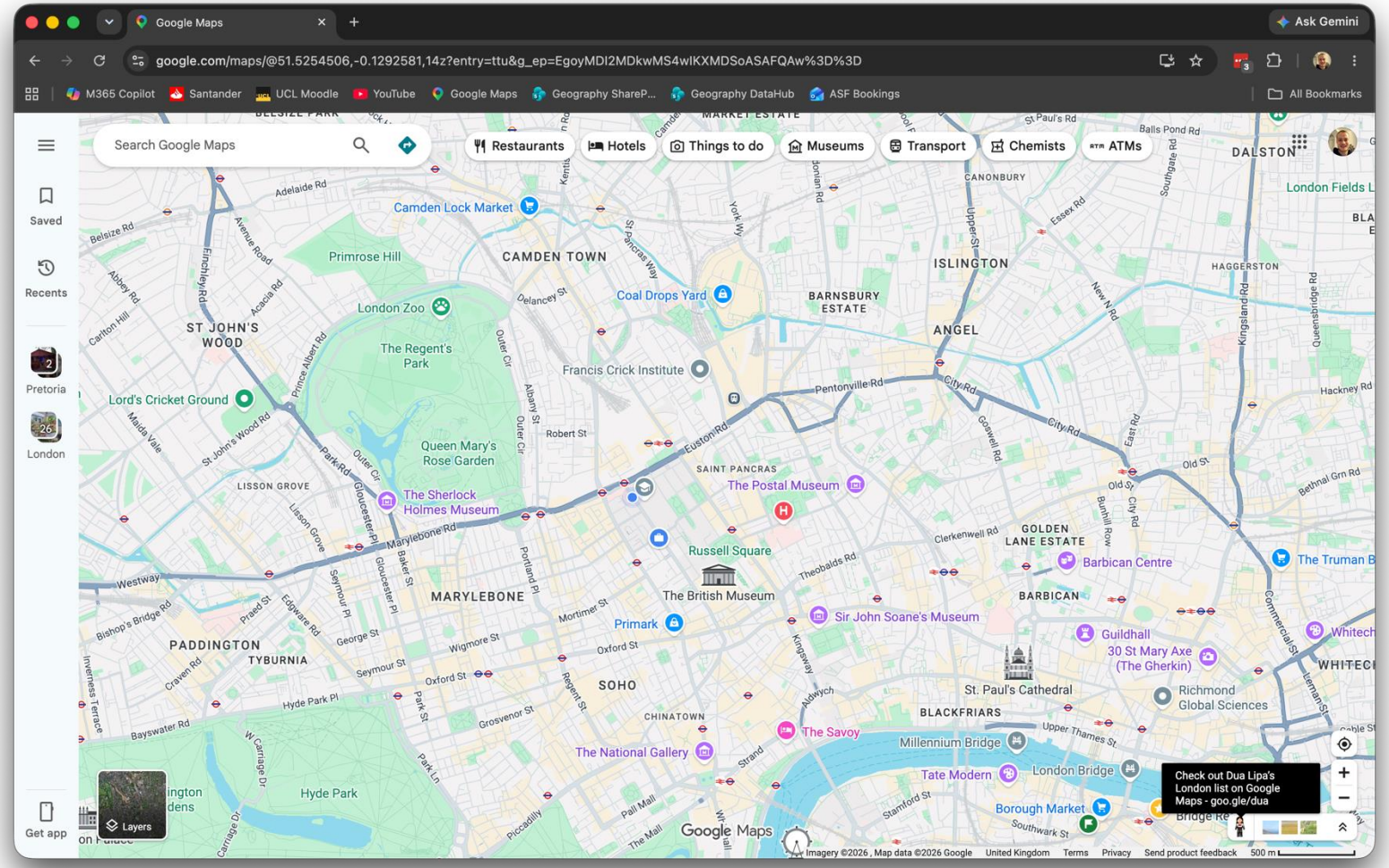
Requirements

- GIScience: Theory
- GISystems: Methods and tools

Digital representation



Digital representation



GIScience

- GIScience relies on representing spatial information in a digital format. Traditionally, geographic information is conveyed in two primary ways:

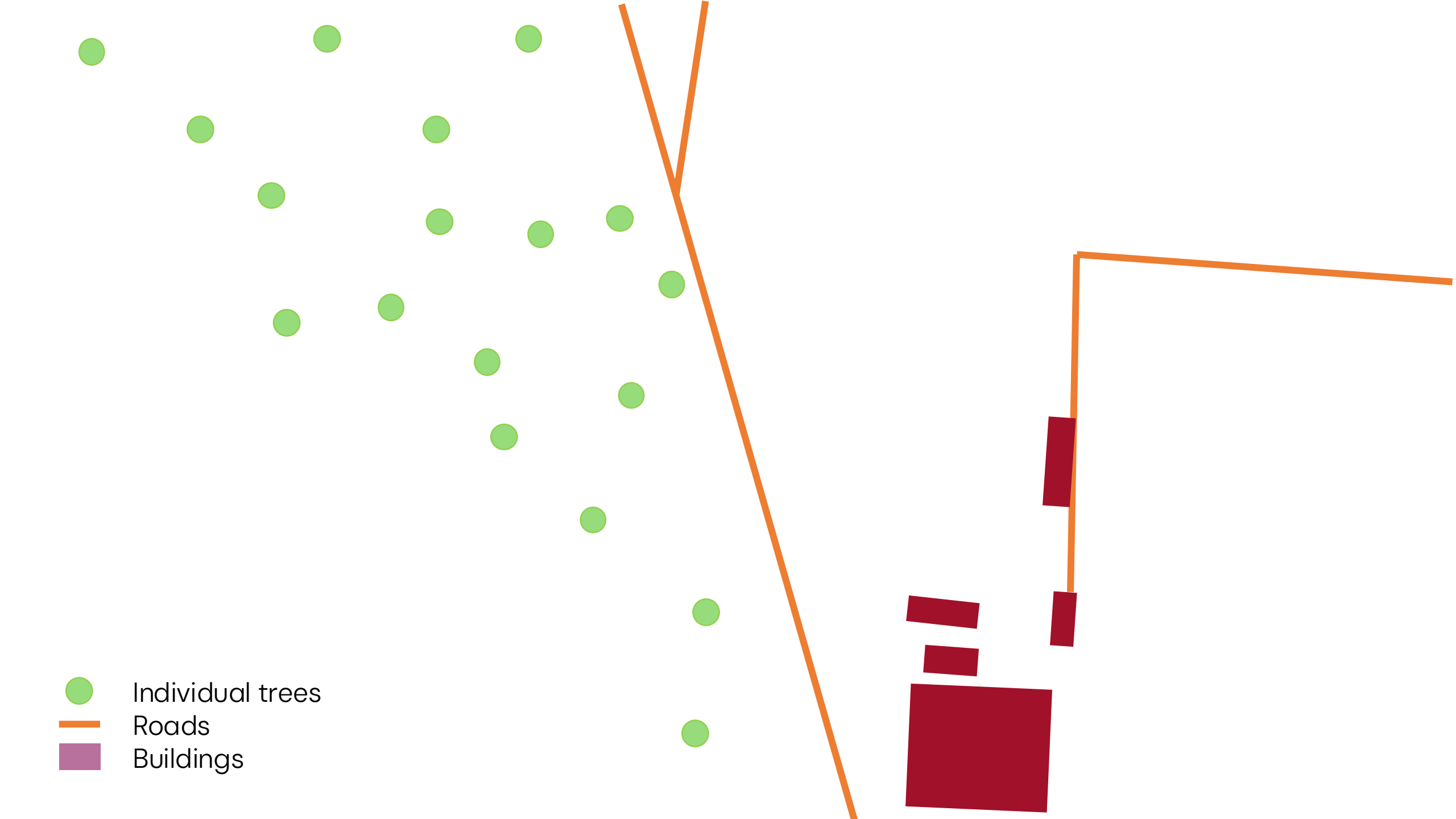
  Vector This method uses a finite set of discrete geometric objects, such as points, lines, and polygons, to represent spatial features.

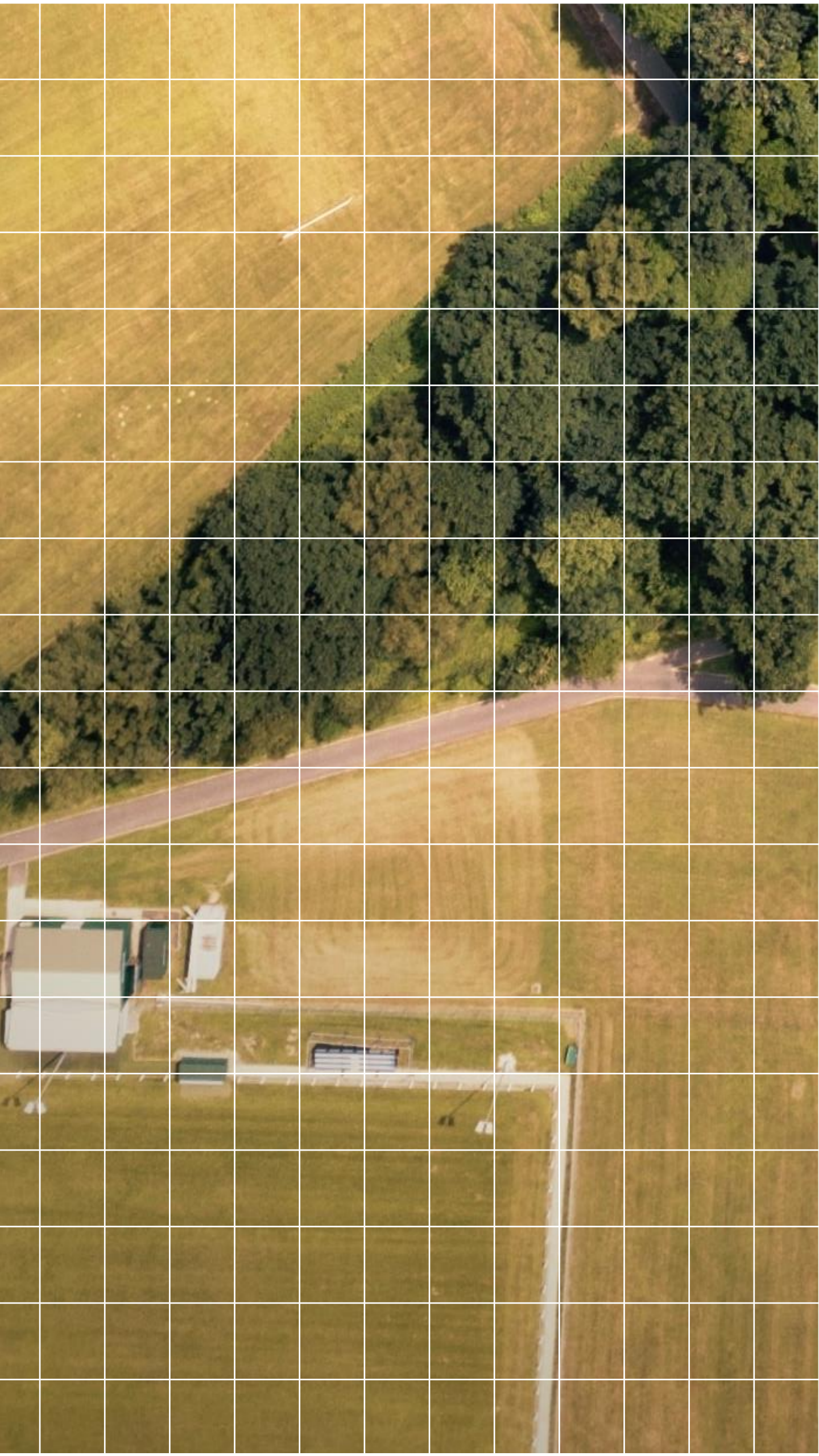
  Raster This approach employs images or grids to represent surfaces, with each cell or pixel holding a value, often indicating attributes like colours or measurements.
- Each feature also has attributes: a row in a table with information about it, such as a road name or a population count.







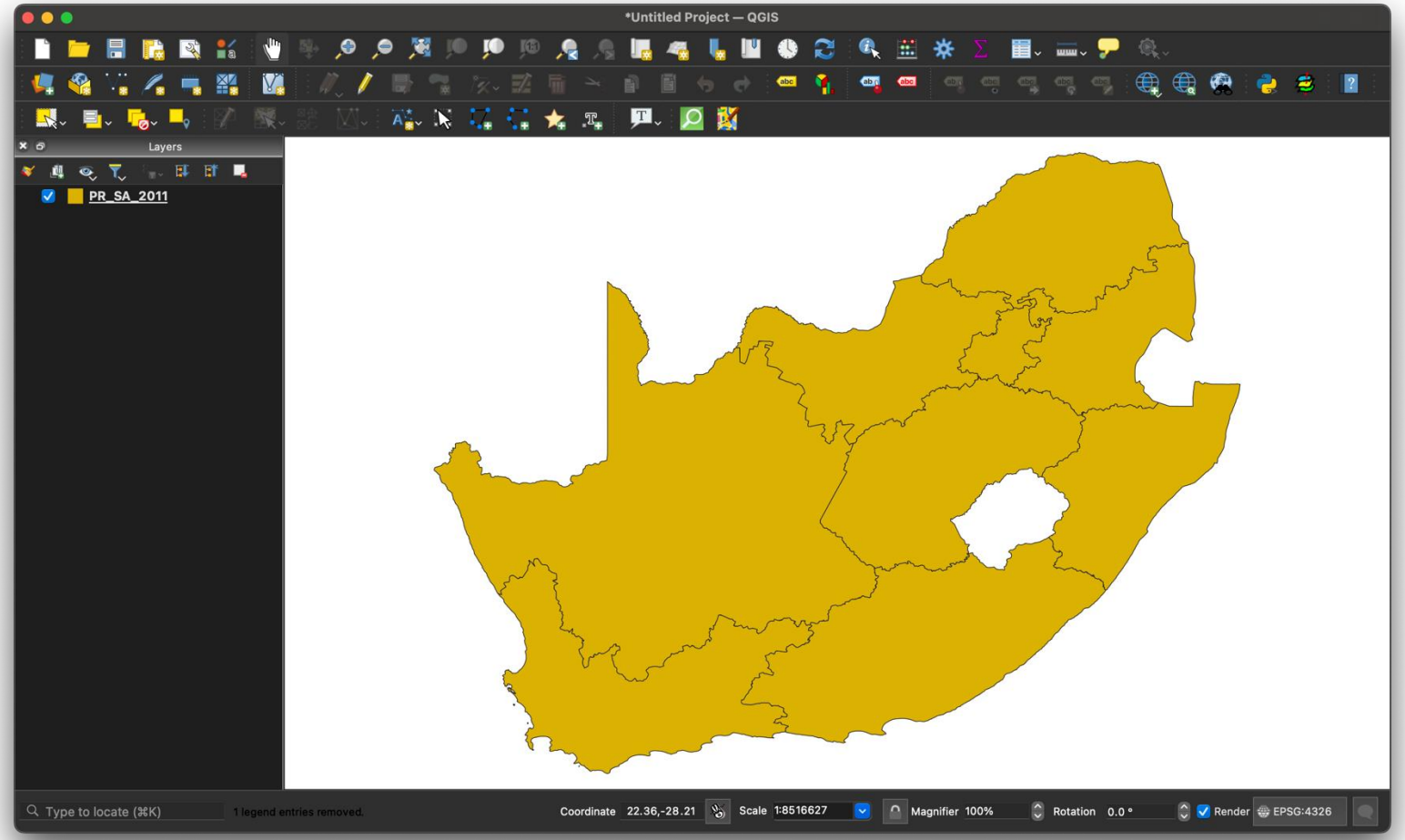

Individual trees
Roads
Buildings



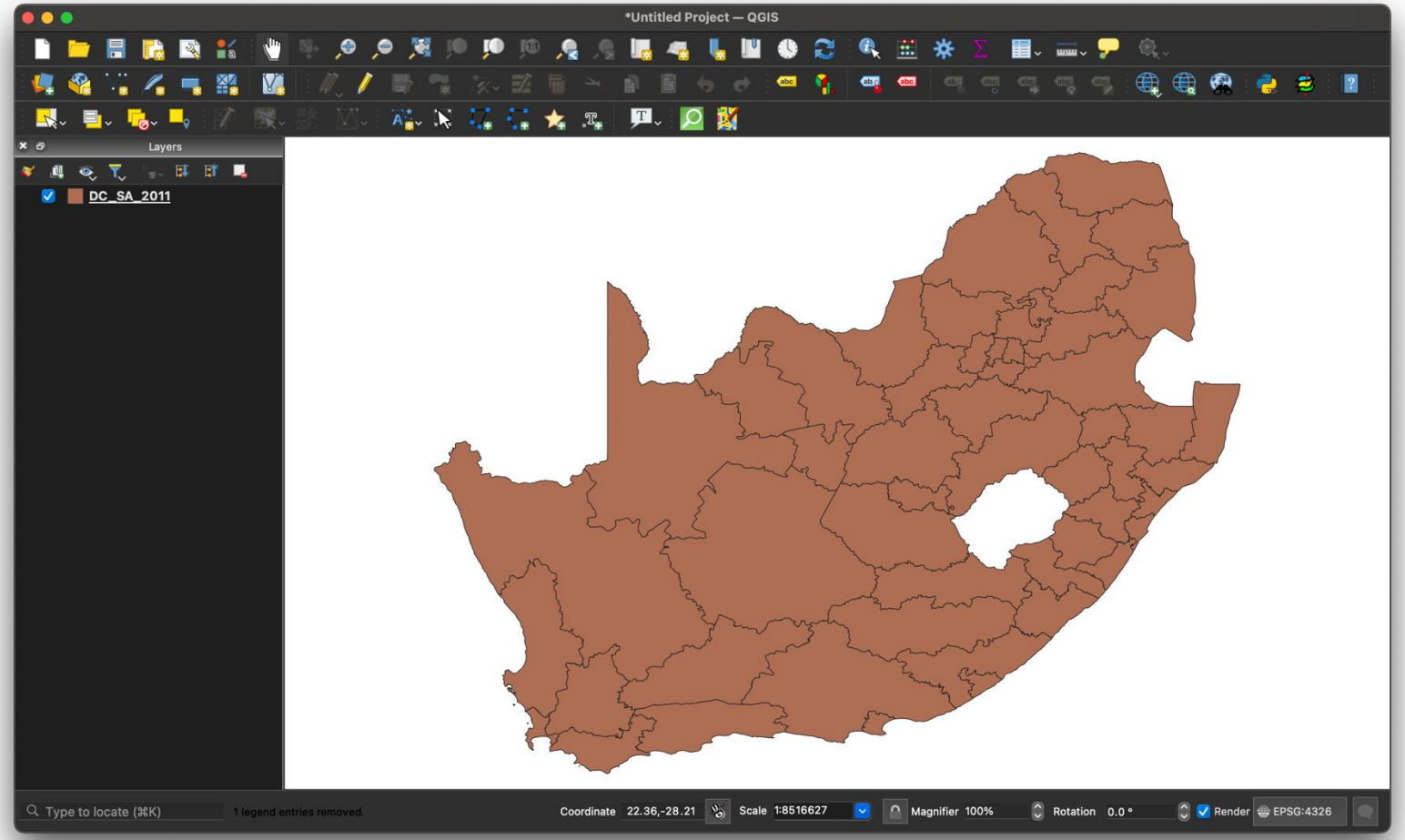


8	9	9	10	0	10	10	10	0	0	0	7	5	3	0	0	0	0	1
8	9	9	10	10	0	10	9	9	0	0	5	3	0	0	0	0	0	0
8	8	9	9	10	0	0	9	8	7	5	0	0	0	1	0	0	0	0
5	8	8	9	10	10	0	9	7	5	0	0	5	5	5	0	0	0	1
3	5	8	9	9	10	0	0	3	0	0	0	5	0	0	1	0	0	2
2	5	8	8	9	9	10	0	0	0	1	5	0	0	0	0	0	0	1
2	4	6	8	8	9	0	0	0	1	5	0	0	5	5	5	0	0	1
0	3	6	8	8	0	0	0	0	5	0	5	5	5	5	5	0	0	0
2	2	5	8	0	0	0	0	0	0	5	5	5	5	5	5	3	0	0
0	2	5	0	0	1	2	3	4	4	4	4	4	4	4	5	0	0	0
0	0	0	0	1	1	1	1	4	4	4	4	4	4	4	5	0	0	0
0	0	1	1	2	2	2	2	3	3	3	3	3	3	3	4	0	3	0
1	1	1	1	2	2	3	3	3	3	1	1	1	1	1	2	3	4	3

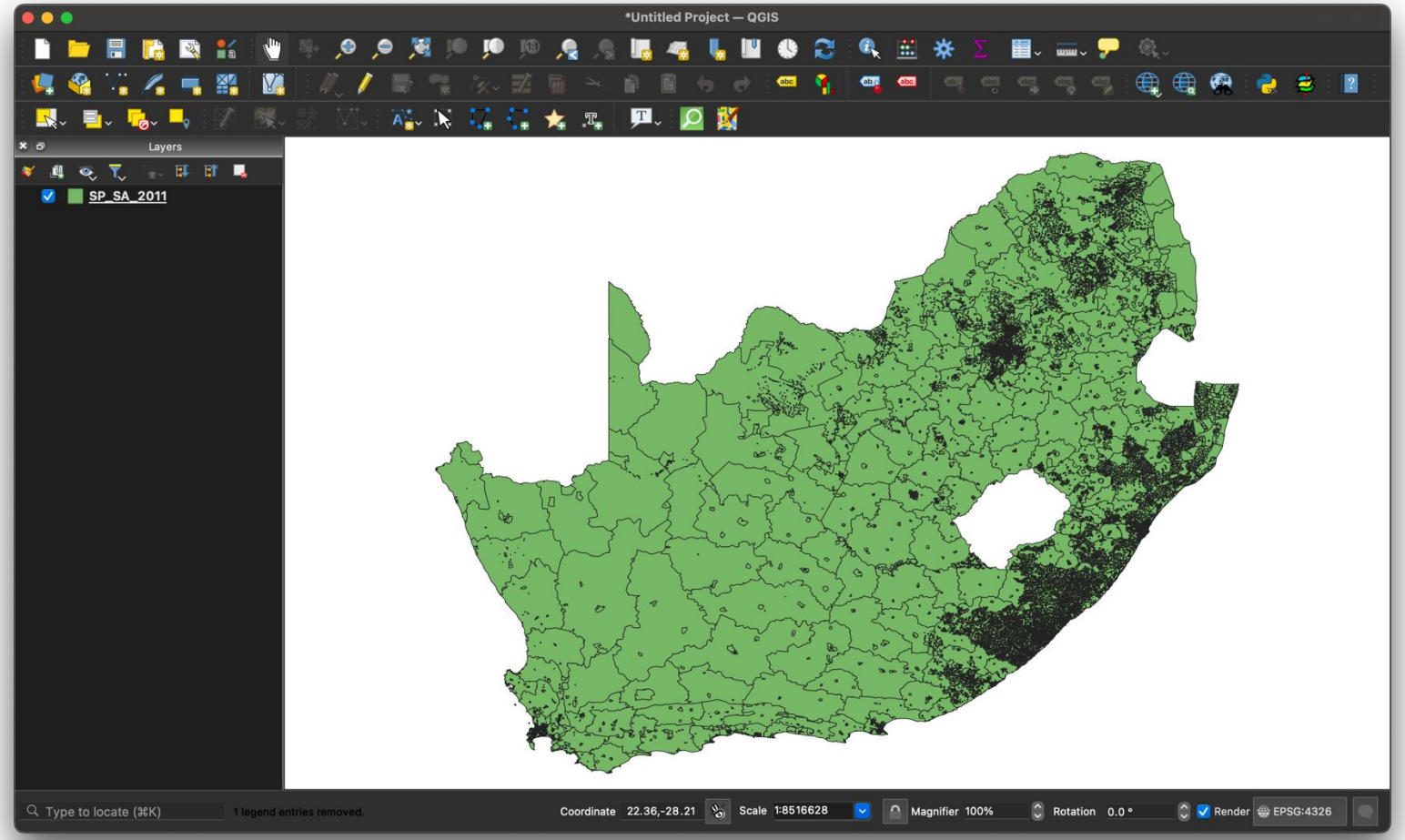
Same place, different levels of detail



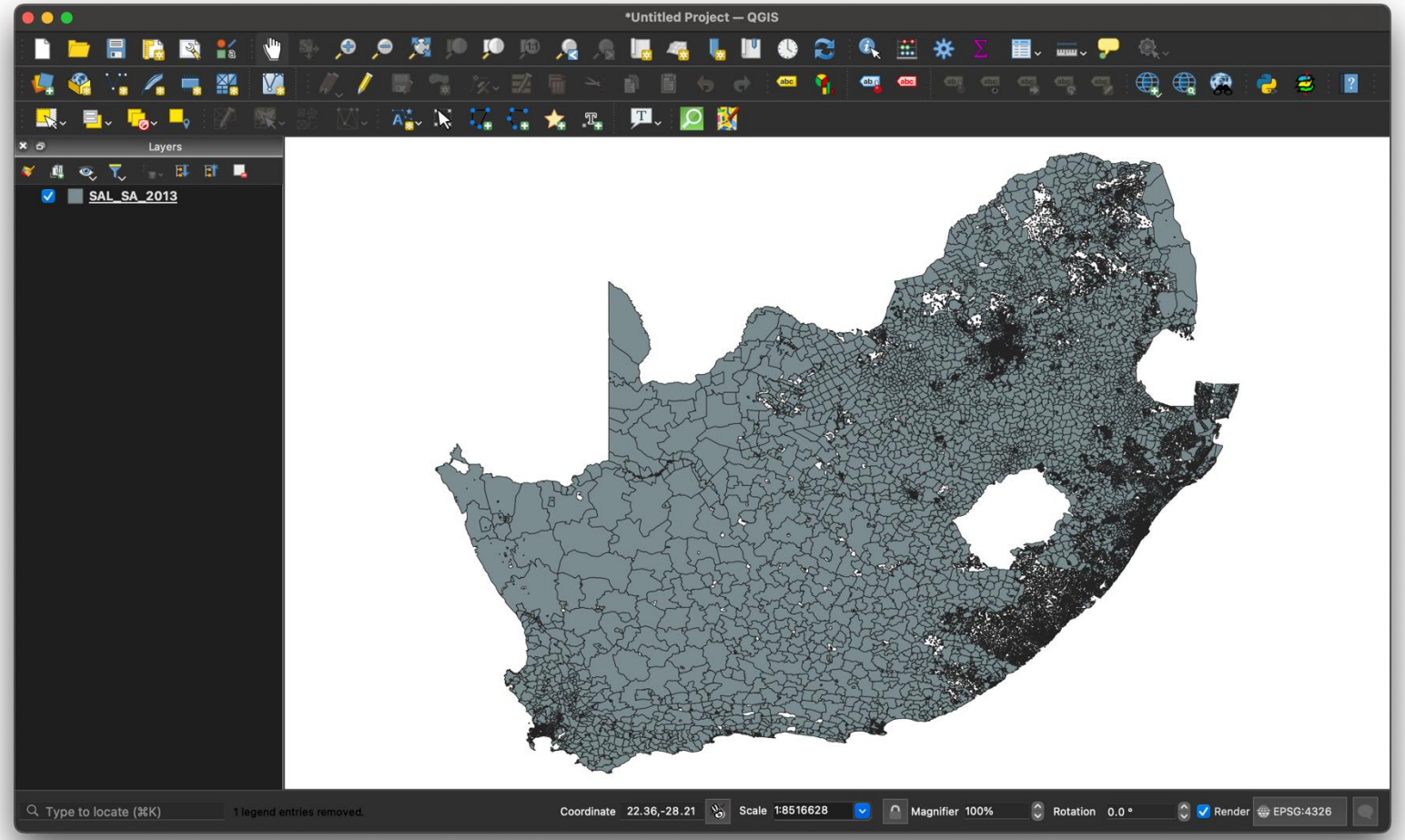
Same place, different levels of detail



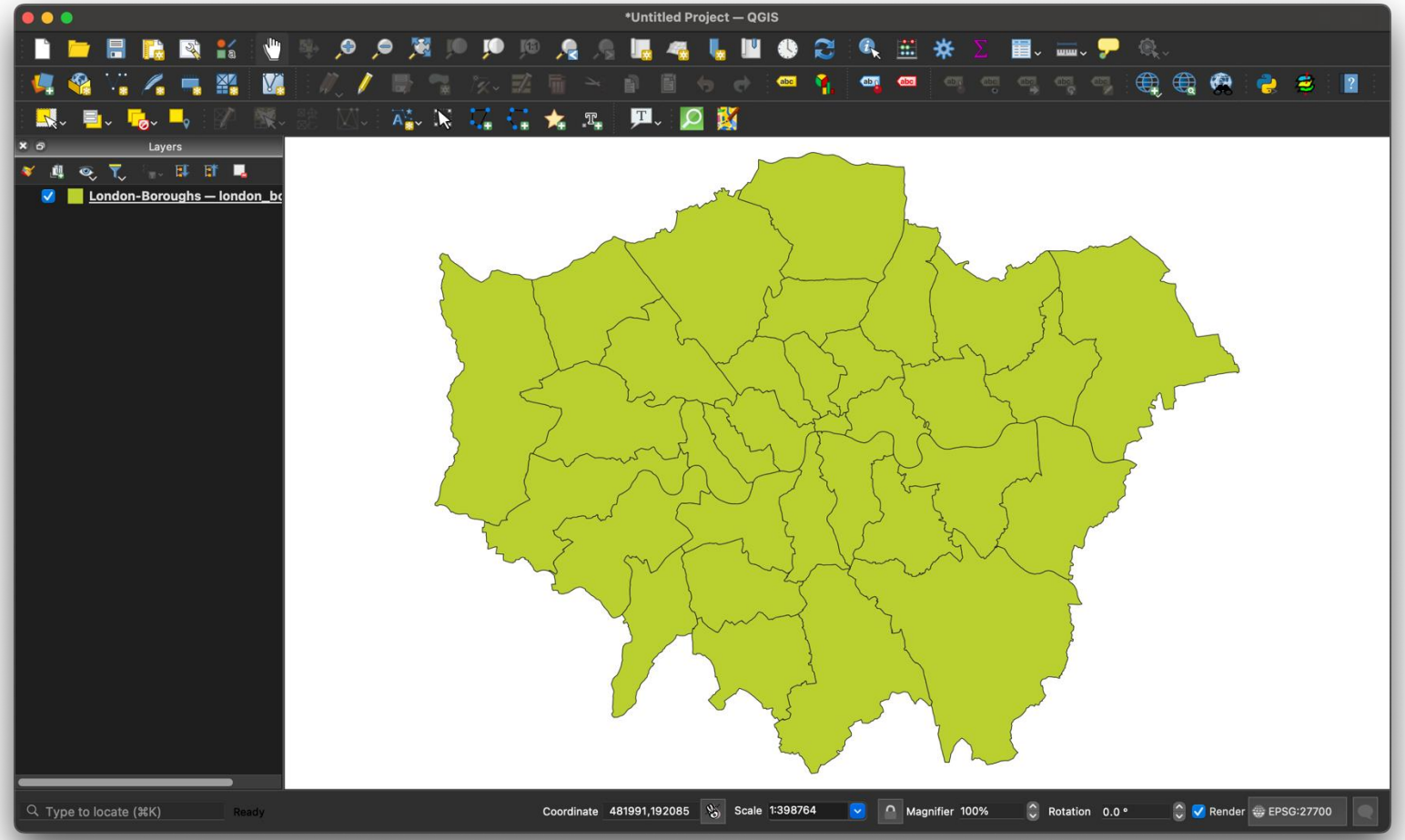
Same place, different levels of detail



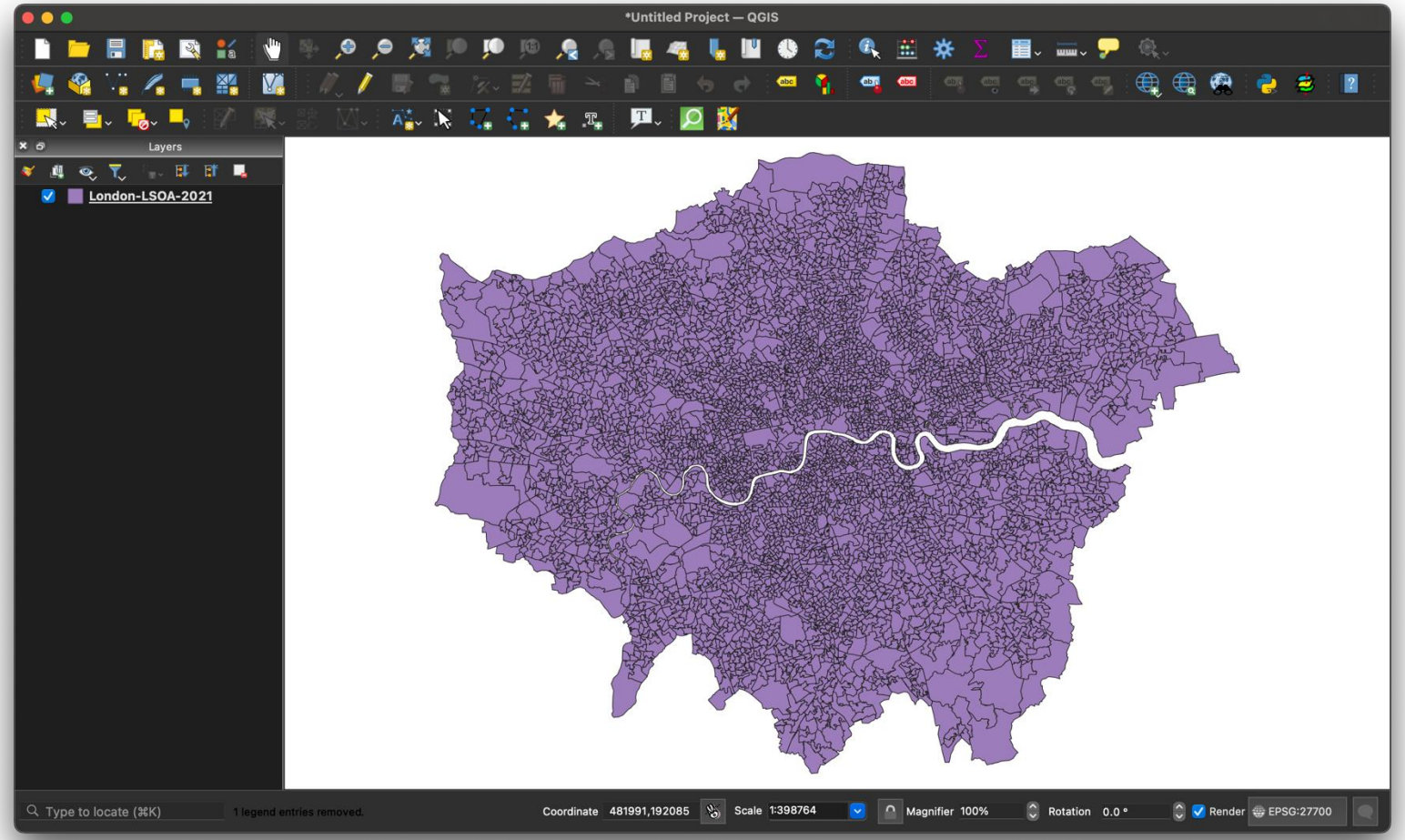
Same place, different levels of detail



Same place, different levels of detail



Same place, different levels of detail



Lower layer Super Output Areas (LSOAs): 1,000 to 3,000 residents each.

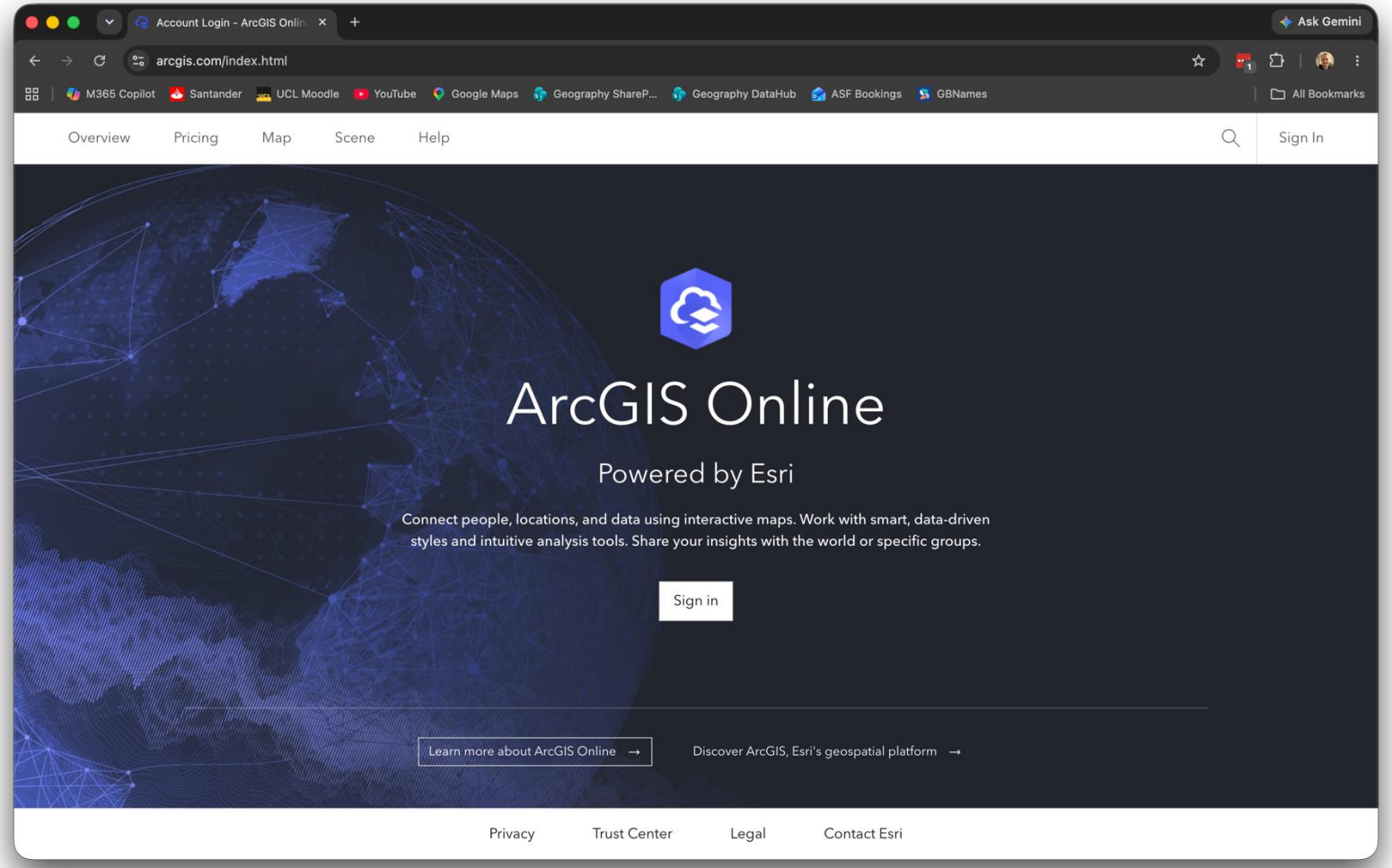
Father of GIS

- Roger F. Tomlinson (1933–2014)
- Ph.D. dissertation: "*The application of electronic computing methods and techniques to the storage, compilation, and assessment of mapped data*" (1974, UCL).
- Conceived the idea of analysing multiple layers of spatial data within a single environment as well as how to represent such spatial data in a digital format.

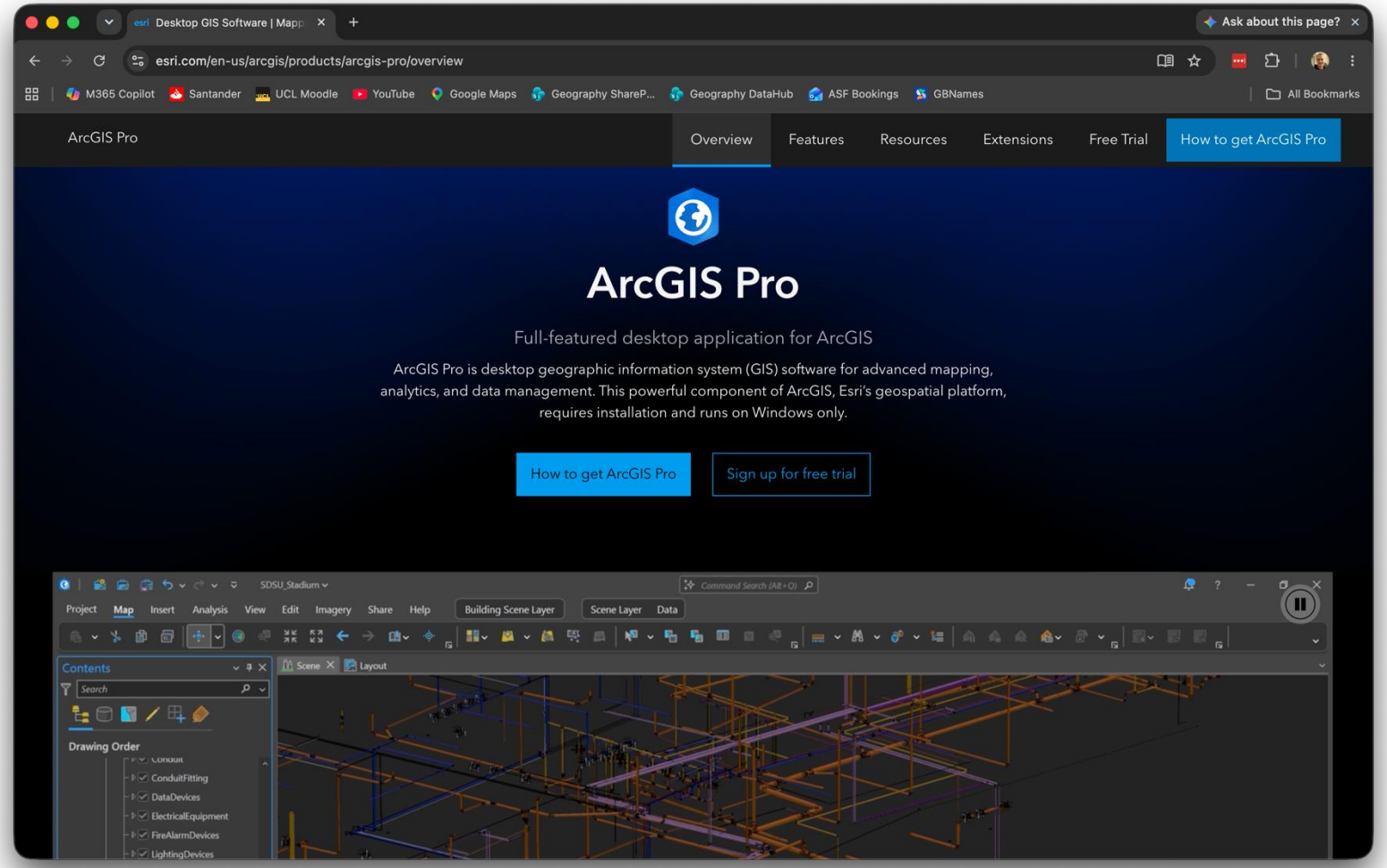
Geographic Information Systems (GIS)

- To effectively handle spatial information, we need tools that can:
  - Collect spatial data
  - Store spatial data
  - Analyse spatial data
  - Present spatial data
- Geographic Information Systems help us to manage spatial data.

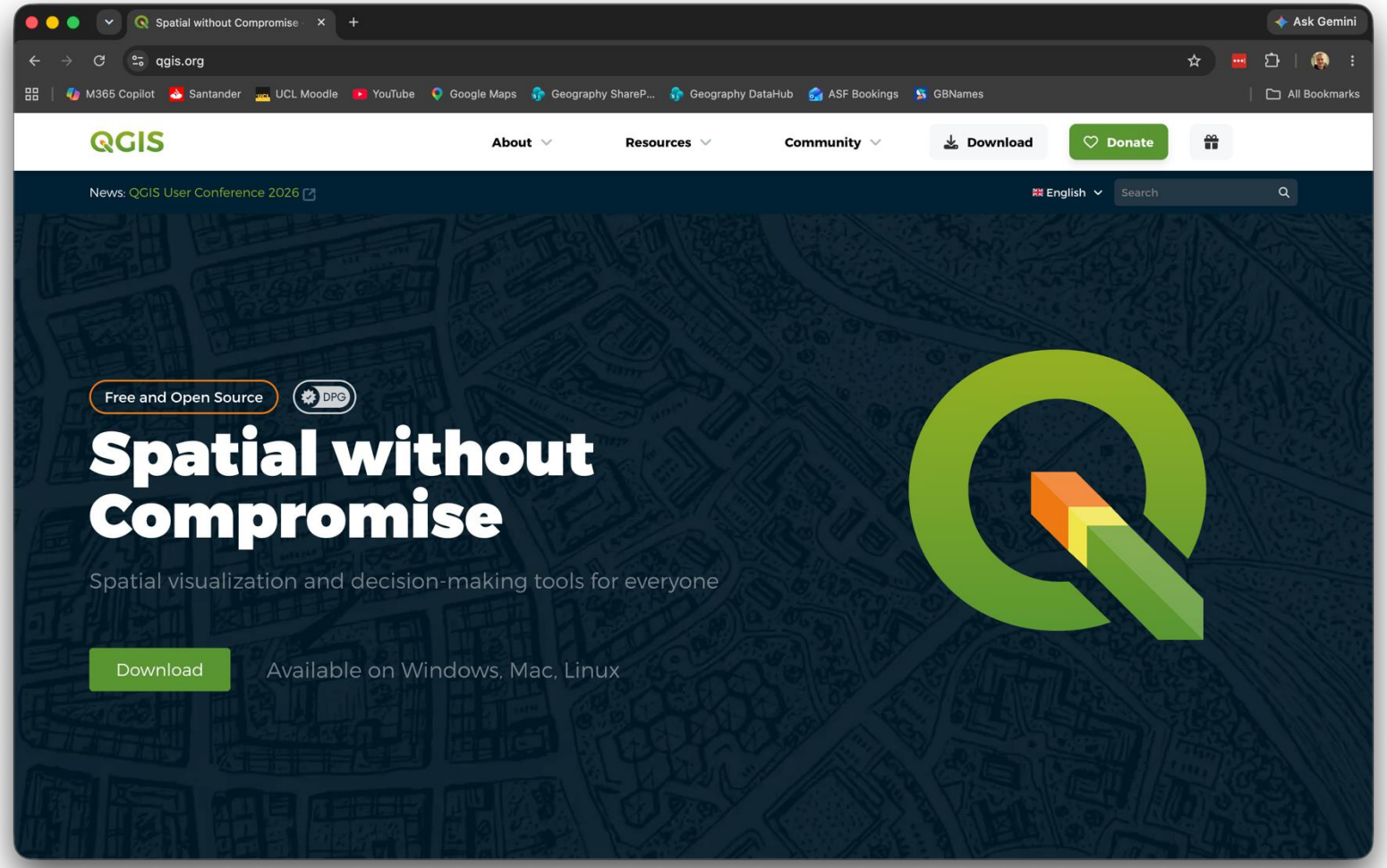
Geographic Information Systems



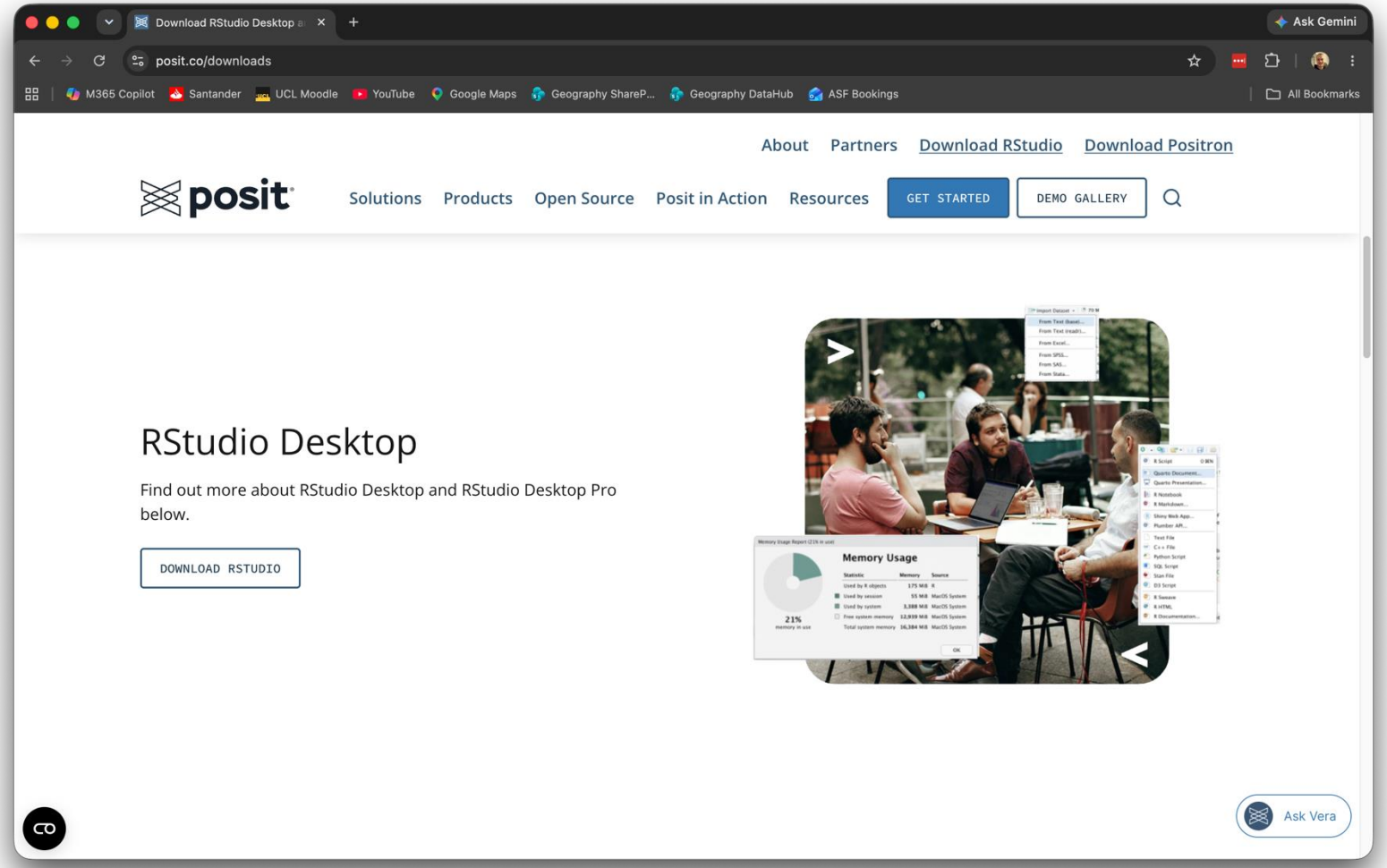
Geographic Information Systems



Geographic Information Systems



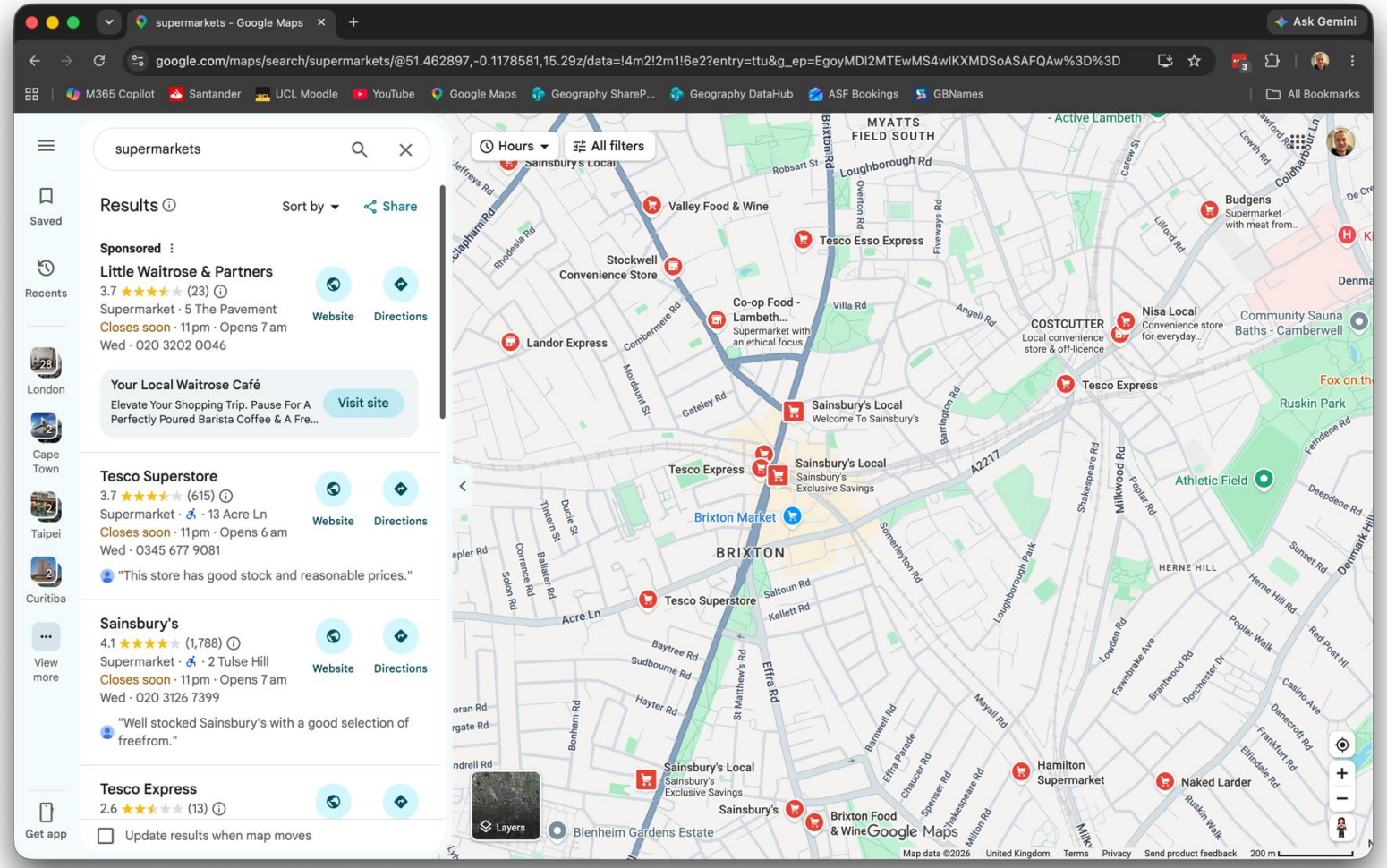
Geographic Information Systems



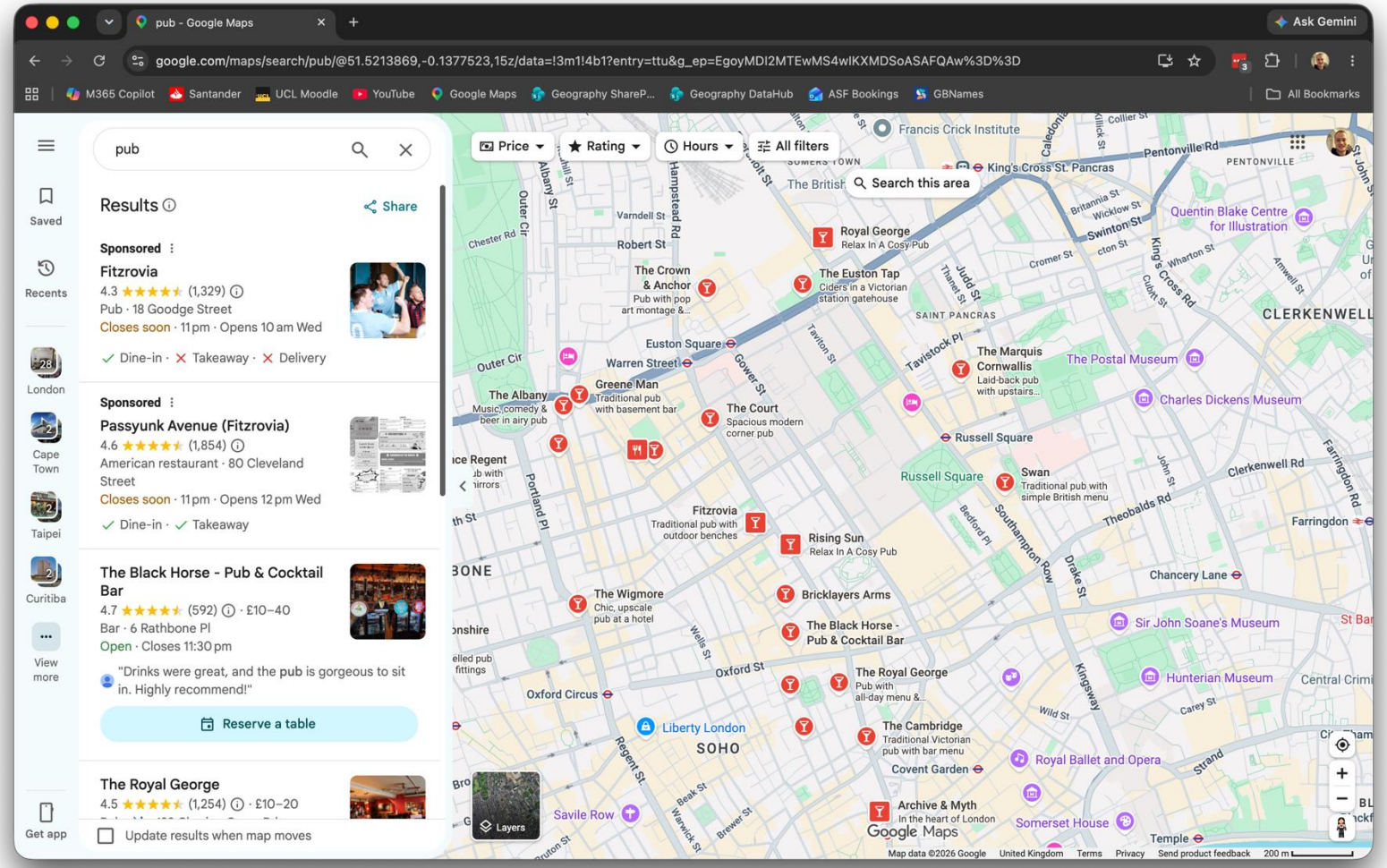
Spatial analysis

- We can use GIScience and GISystems to convert spatial data into actionable information by quantifying elements such as distributions and spatial processes.
- In plain words: finding patterns in where things are and how places compare.

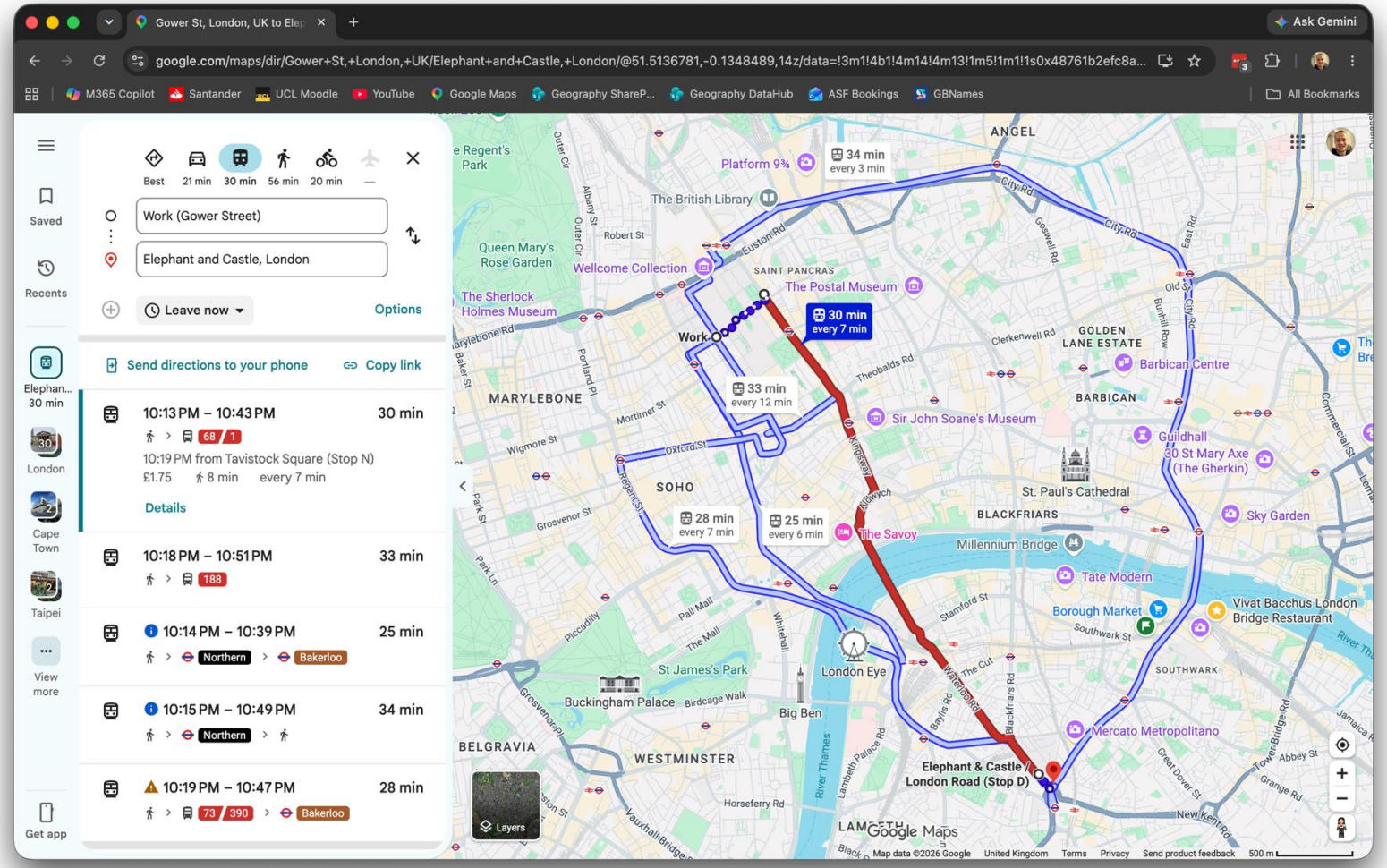
Spatial analysis: finding locations



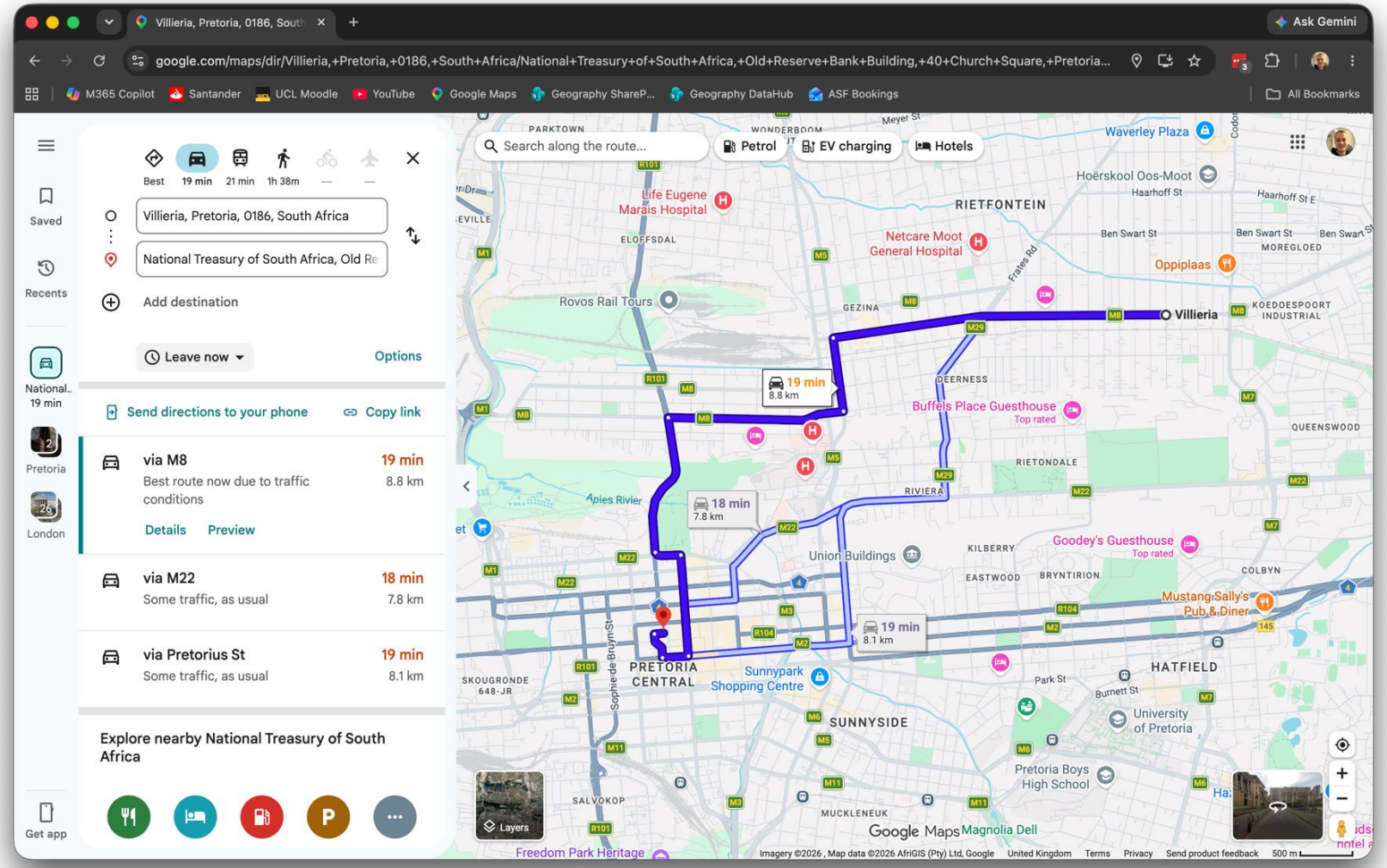
Spatial analysis: finding locations



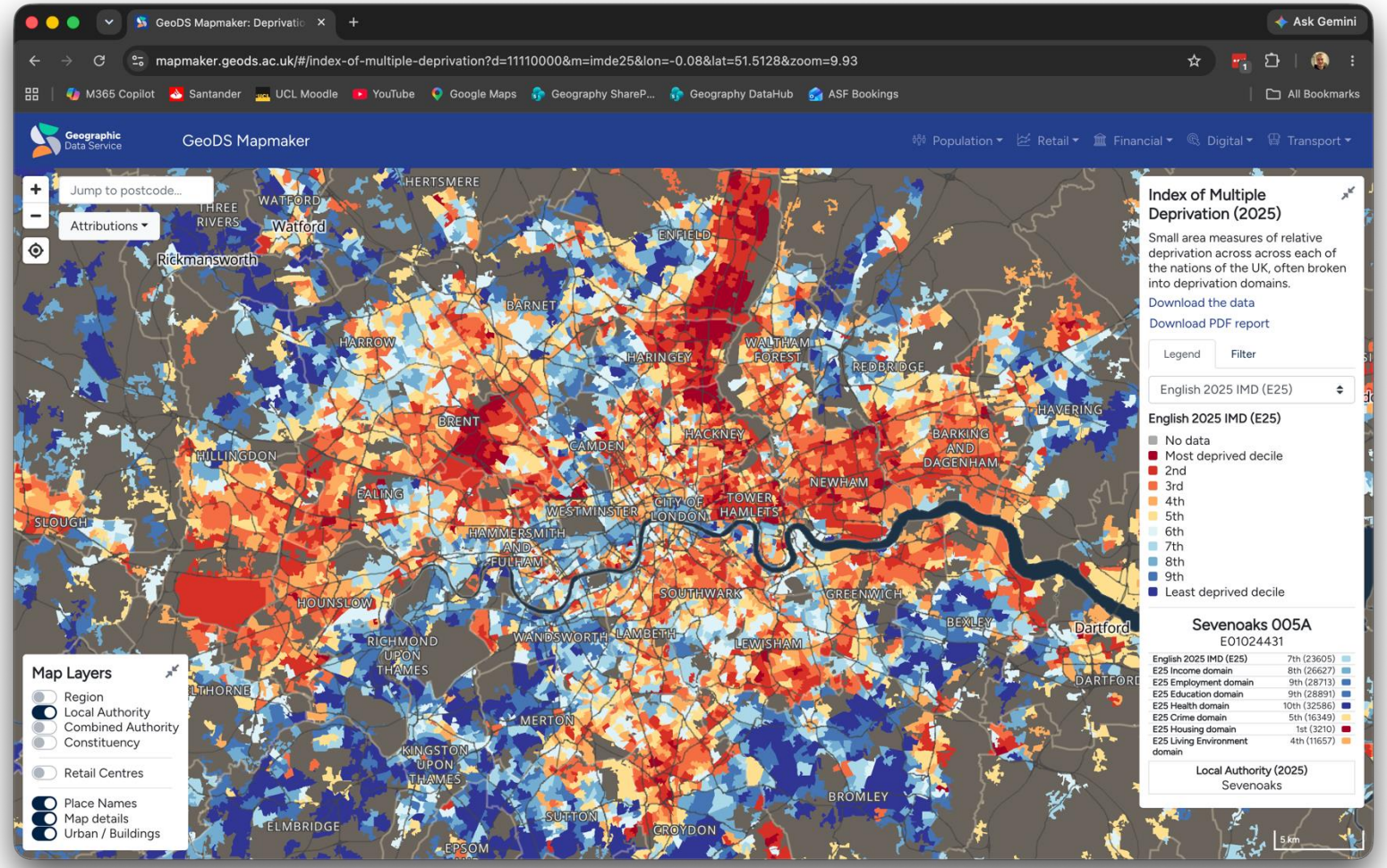
Spatial analysis: finding routes



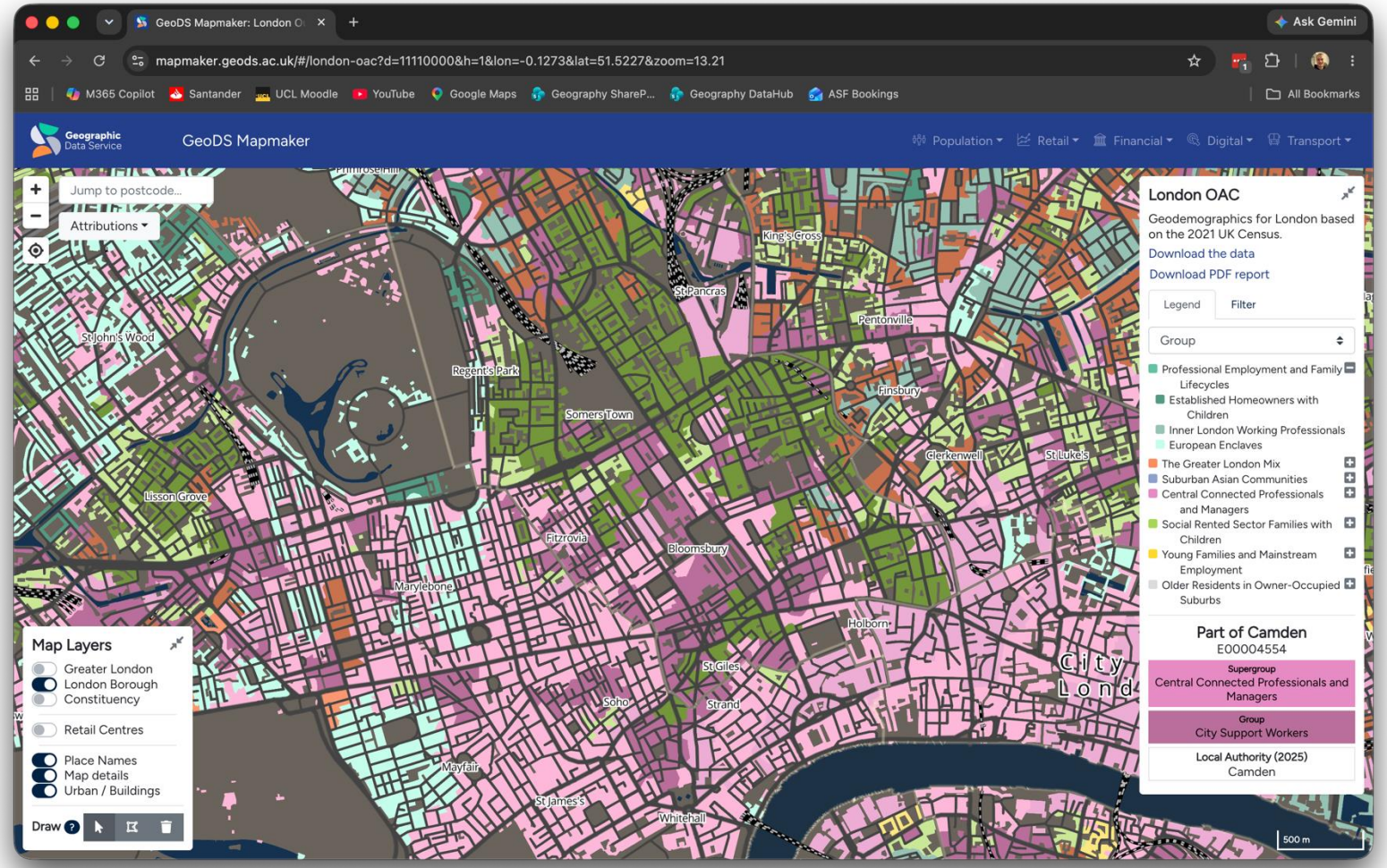
Spatial analysis: finding routes



Spatial analysis: comparing places



Spatial analysis: comparing places



File formats: Geopackage

- A GeoPackage is an open, standards-based, platform-independent, portable, self-describing, compact format for transferring geospatial data.
- It stores spatial data layers in a single file, based on an SQLite database.
- How to spot in the wild: .gpkg

File formats: Shapefile

- `.shp` Contains the feature geometry.
- `.shx` Index file which stores the position of the feature's ID in the `.shp` file.
- `.dbf` Stores all attribute information associated with the records.
- `.prj` Contains the coordinate system information and projection.
- `.xml` General metadata.
- `.cpg` Encoding information.
- `.sbn` *O*ptimisation file for spatial queries.

File formats: Shapefile



Coordinate reference systems

- A coordinate reference system (CRS) defines how coordinates relate to real places on the Earth.
- British National Grid (EPSG:27700): metres, used for Great Britain.
- WGS 84 (EPSG:4326): latitude and longitude in degrees, used by GPS.
- Layers in different CRSs can end up in the wrong place.

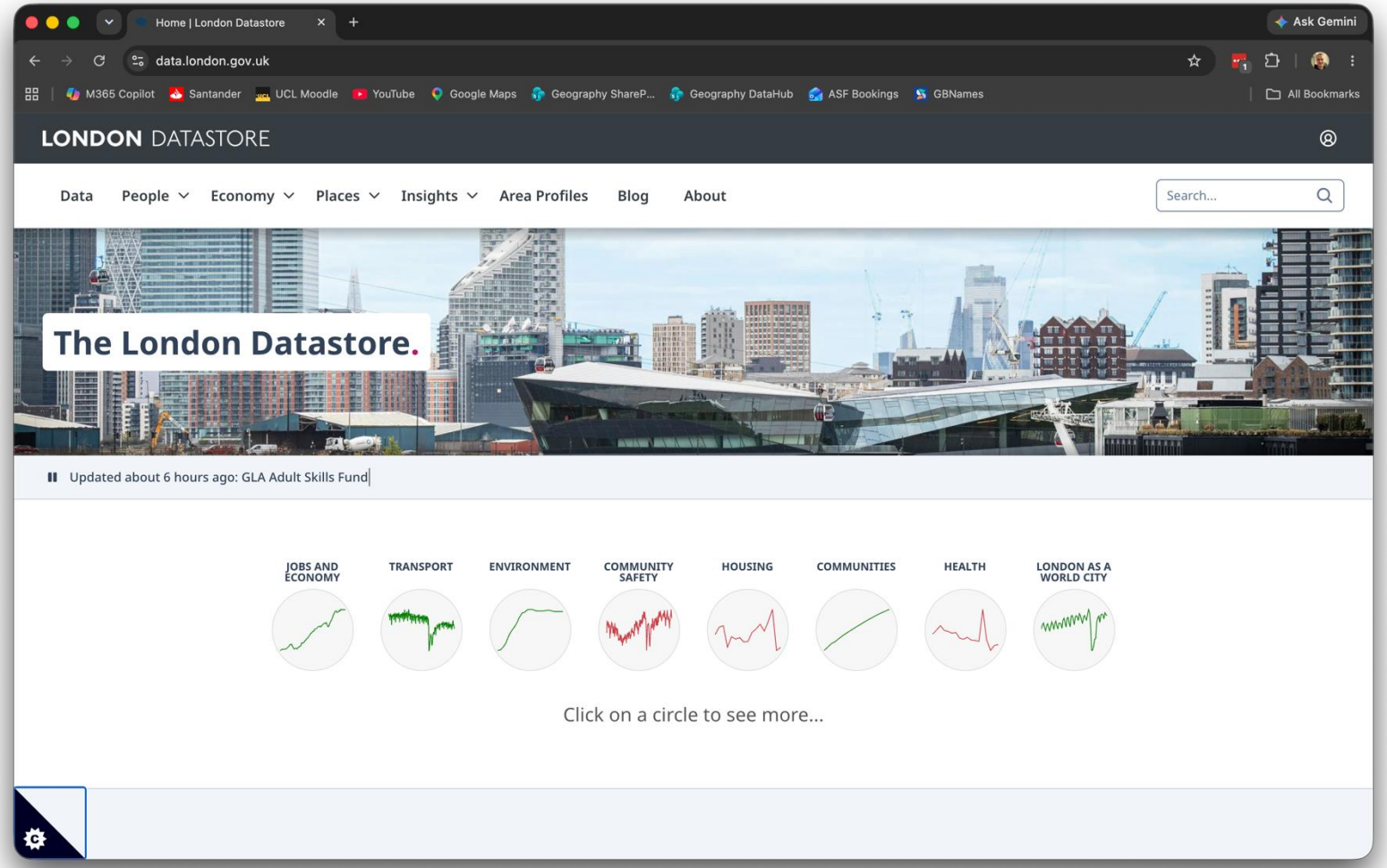
Mentimeter

- Go to www.menti.com.
- Use code: XXXX XXXX

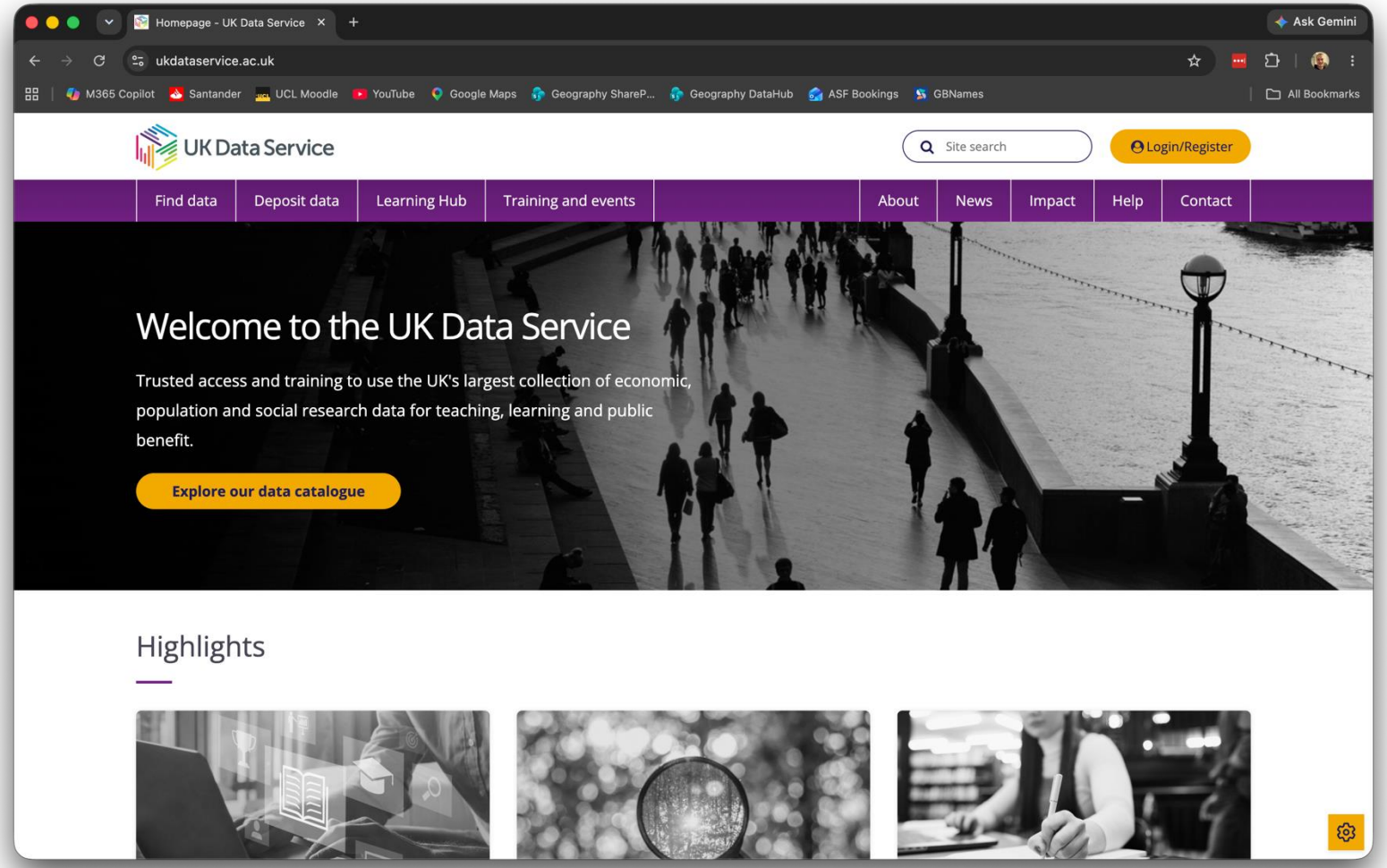
Break

Data sources

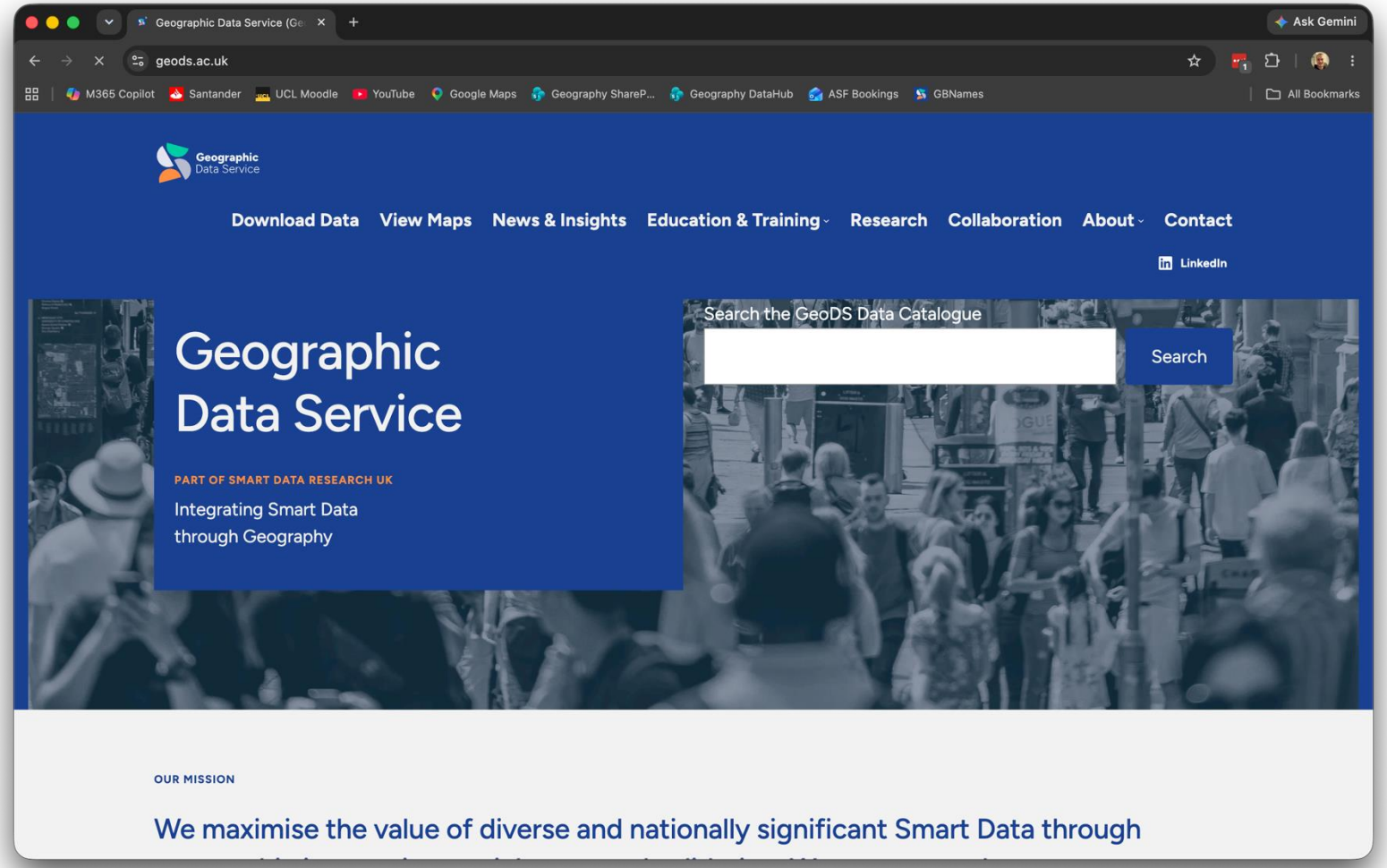
London Datastore



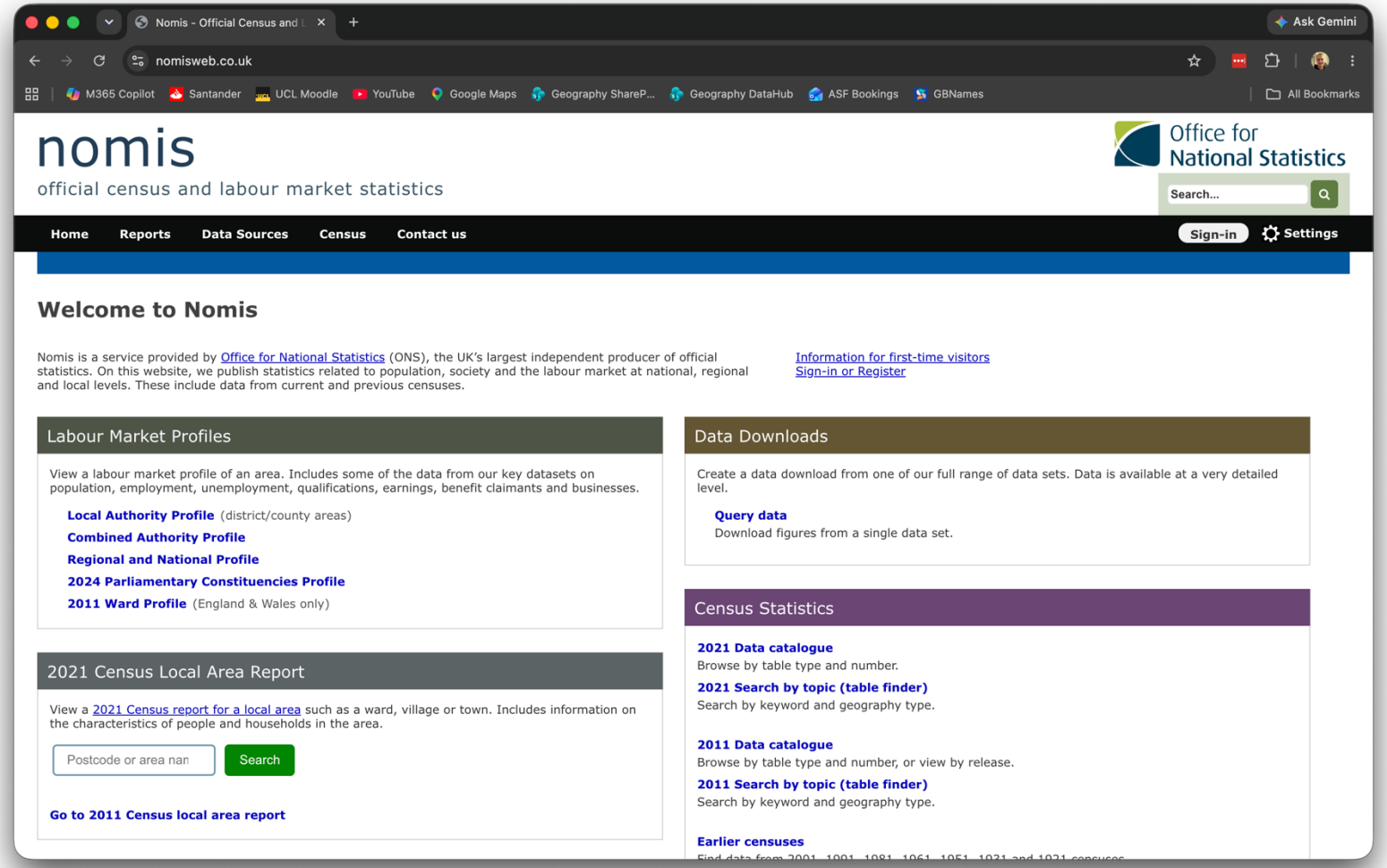
UK Data Service



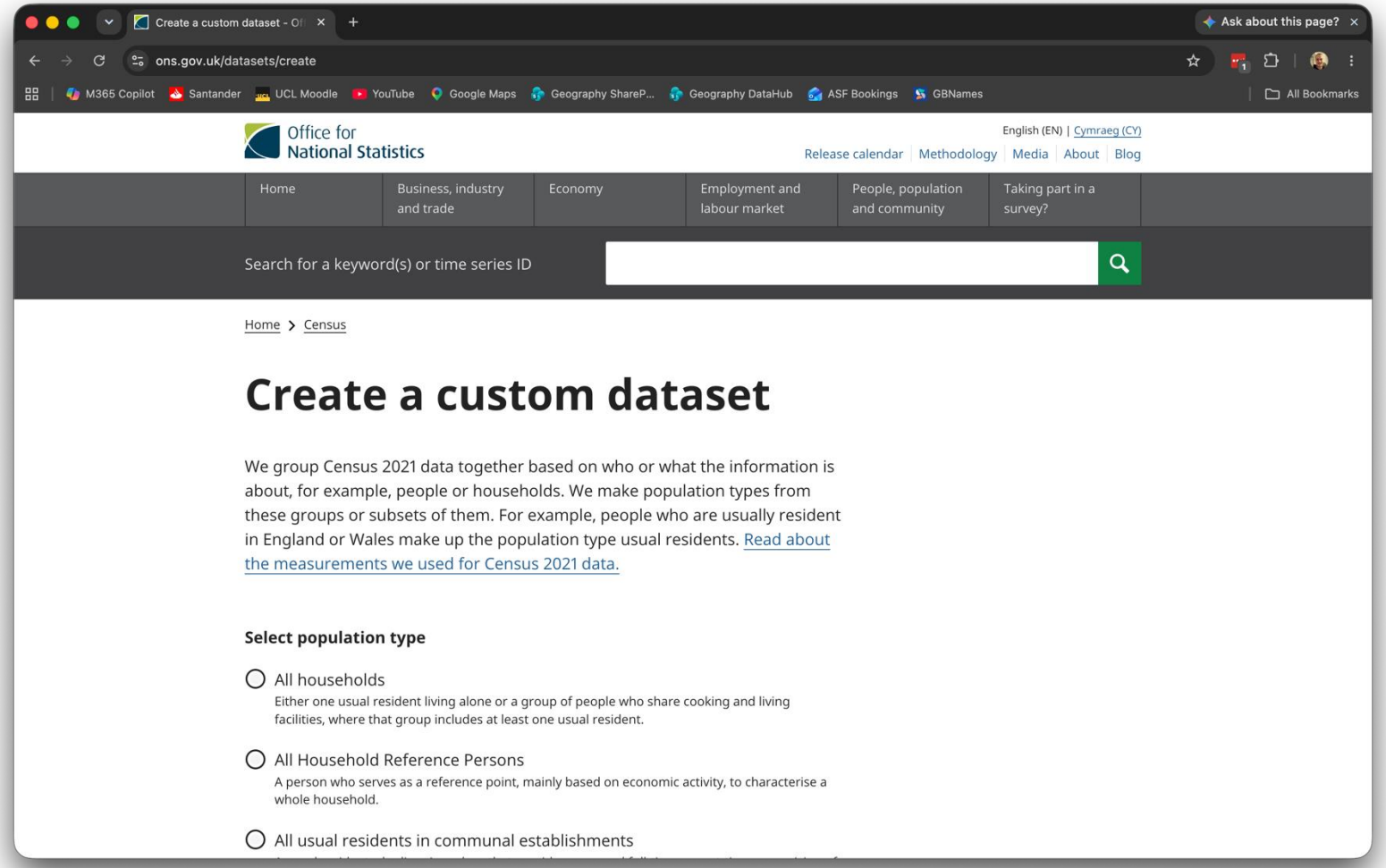
Geographic Data Service



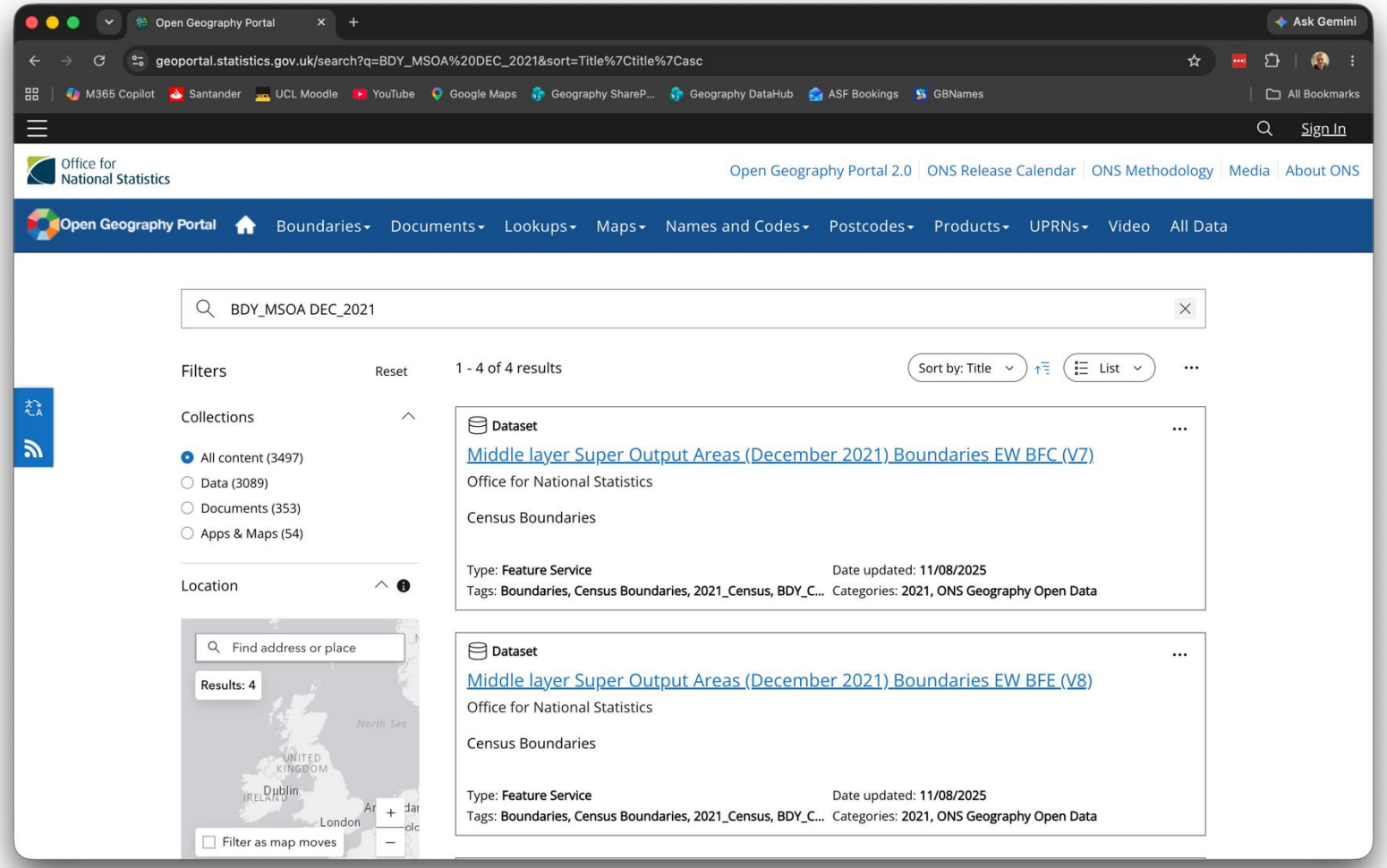
nomis

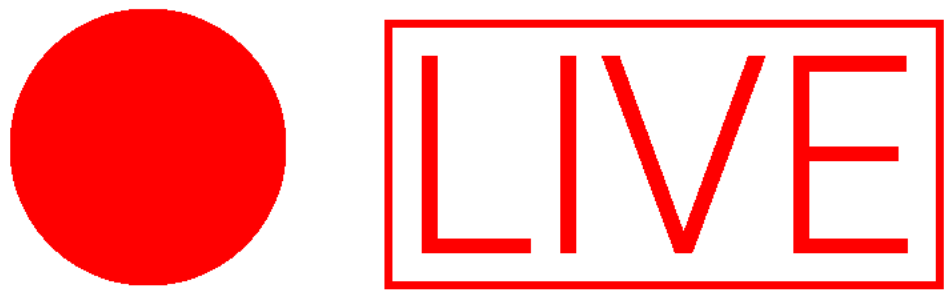

Nomis - Official Census and
Ask Gemini
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M365 Copilot
Santander
UCL Moodle
YouTube
Google Maps
Geography ShareP...
Geography DataHub
ASF Bookings
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View a labour market profile of an area. Includes some of the data from our key datasets on population, employment, unemployment, qualifications, earnings, benefit claimants and businesses.
Local Authority Profile (district/county areas)
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Regional and National Profile
2024 Parliamentary Constituencies Profile
2011 Ward Profile (England & Wales only)
2021 Census Local Area Report
View a 2021 Census report for a local area such as a ward, village or town. Includes information on the characteristics of people and households in the area.
Postcode or area nan
Search
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Data Downloads
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2011 Data catalogue
Browse by table type and number, or view by release.
2011 Search by topic (table finder)
Search by keyword and geography type.
Earlier censuses

Office for National Statistics



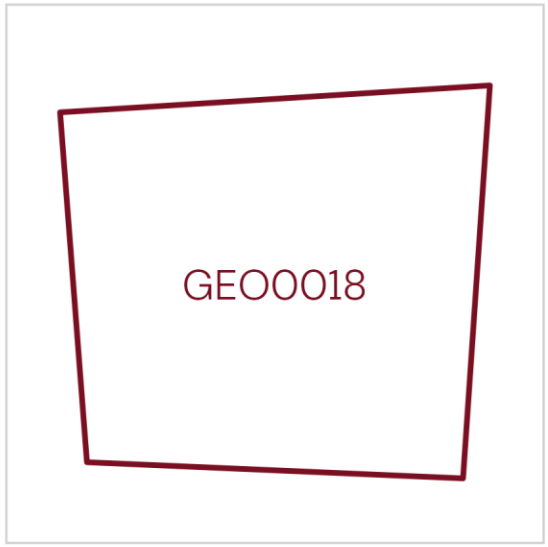
Open Geography Portal




LIVE

Attribute joins

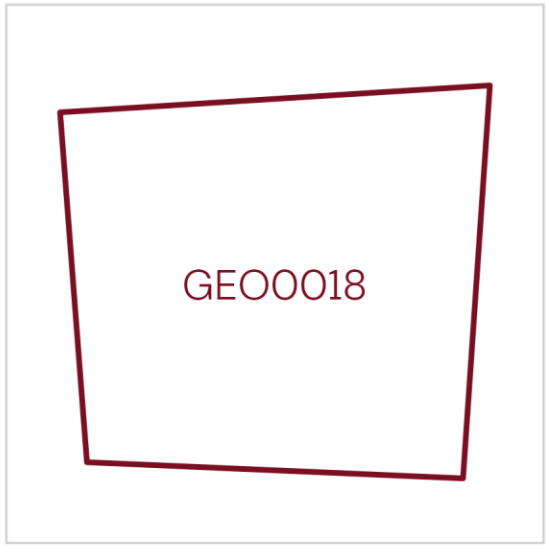
Attribute join: match rows by a shared ID



Features

FeatureID	GeoID
0050	GEO0018

Attribute join: match rows by a shared ID



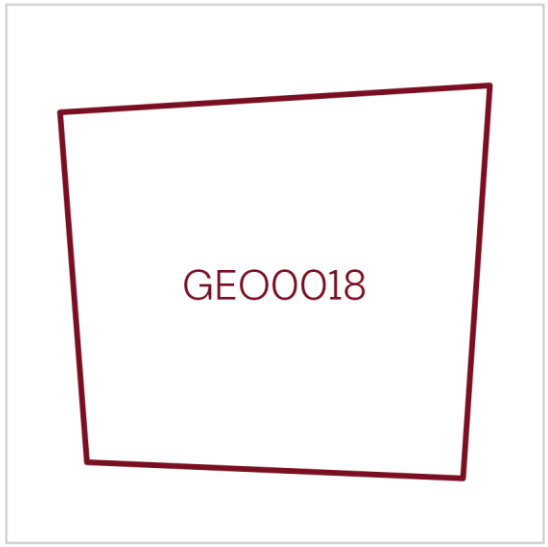
Features

FeatureID	GeoID
0050	GEO0018

Outside data

GeoID	Population
GEO0018	540
GEO0019	320

Attribute join: match rows by a shared ID



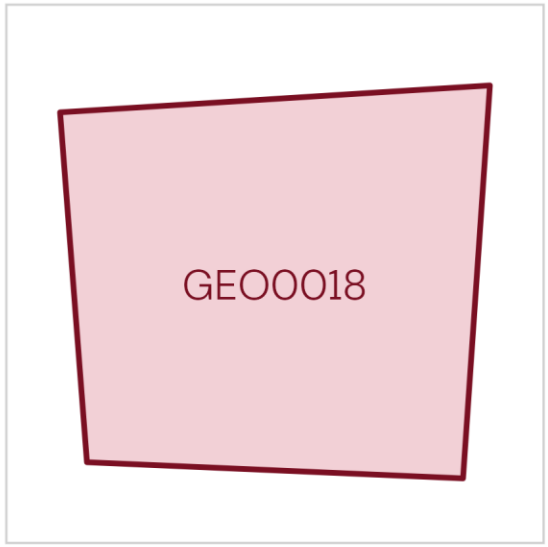
Features

FeatureID	GeoID
0050	GEO0018

Outside data

GeoID	Population
GEO0018	540
GEO0019	320

Attribute join: match rows by a shared ID



Features with the outside data added

FeatureID	GeoID	Population
0050	GEO0018	540

Left join: keep every row of the first table

Table 1

id	a
1	a1
2	a2

Table 2

id	b
1	b1
3	b3
4	b4

Left join

id	a	b
1	a1	b1
2	a2	NA

id 2 has no match in Table 2, so its value from Table 2 is NA.

In QGIS every polygon stays; unmatched polygons get empty values.

Inner join: keep only rows found in both

Table 1

id	a
1	a1
2	a2

Table 2

id	b
1	b1
3	b3
4	b4

Inner join

id	a	b
1	a1	b1

Only id 1 appears in both tables, so only that row is kept.

Outer join: keep every row from both tables

Table 1

id	a
1	a1
2	a2

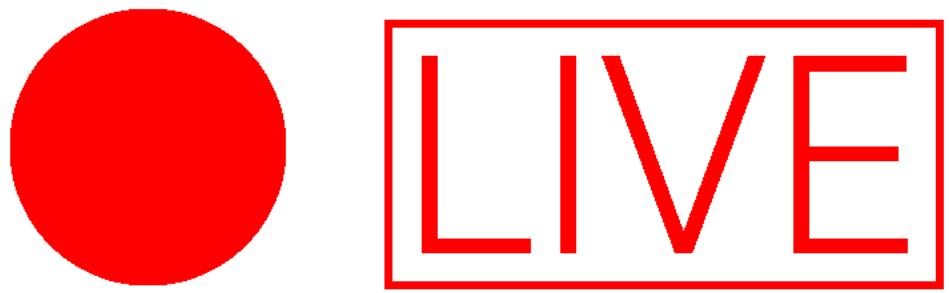
Table 2

id	b
1	b1
3	b3
4	b4

Outer join

id	a	b
1	a1	b1
2	a2	NA
3	NA	b3
4	NA	b4

Every id from either table is kept; a missing match is NA.

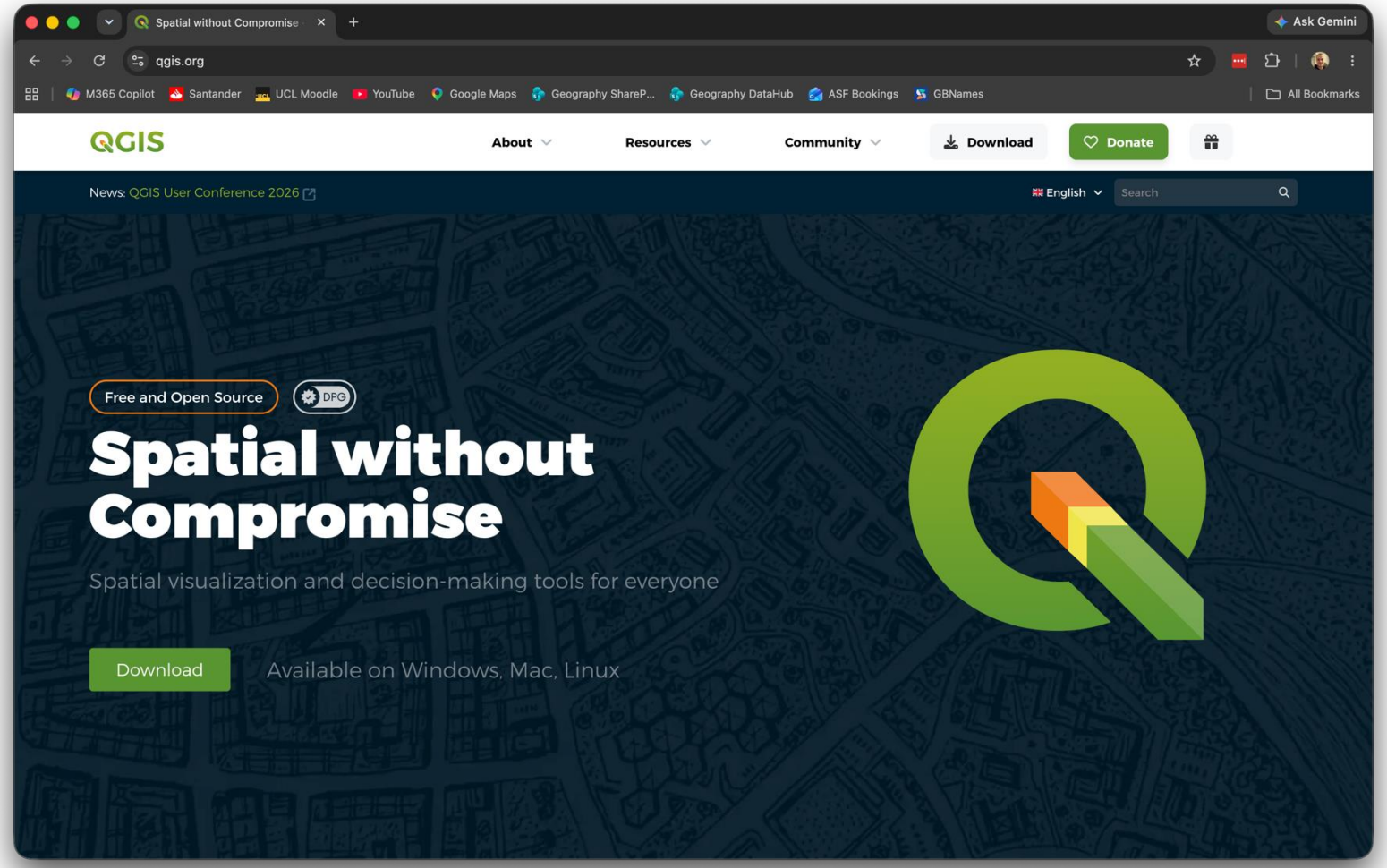

LIVE

Summary

Summary

- Many challenges and problems are inherently geographical.
- Spatial data visualisation is an essential part of (quantitative) human geography for exploring and communicating spatial patterns.
- A wide range of secondary data sources is available, covering topics such as demographics, environment, and socio-economic trends.
- Spatial data is represented as vector (points, lines, polygons) or raster (grid cells).
- An attribute join links a table to spatial features through a shared unique identifier.

Summary


Spatial without Compromise
Ask Gemini
qgis.org
M365 Copilot
Santander
UCL Moodle
YouTube
Google Maps
Geography ShareP...
Geography DataHub
ASF Bookings
GBNames
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Spatial visualization and decision-making tools for everyone
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Available on Windows, Mac, Linux

Assessment

To explore a geographic theme (e.g. greenspace, labour, mobility) in the context of London, submitted as a 1,500-word report (50% of module grade):

1. Outline a research question and hypothesis (200–250 words).
2. Obtain data and identify appropriate methods to use (200–250 words).
3. Present and interpret results (700–800 words).
4. Briefly evaluate your approach (200–300 words).

Deadline Thursday 14 January 2027 at 13h00.

Questions

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